

MIST_Autobots Team Notebook

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1	000	
1.1	Remember [9 lines] - 575b8cdd	
# Don't stuck in one approach		
# Bruteforce is sometimes the best way to go		
# Reset global variables		
# Check for overflow		
# Read the problem carefully		
# Break the problem into smaller parts (go step by step)		
# Check for negative numbers, zero, one or small cases		
# For problems like making string A to B, try to work backwards from B to A or it may be easier to make both A and B to some C		
# L > R can occur		
2	111.Setup	
2.1	FAST_IO [9 lines] - 3f992170	
// C++		
#pragma GCC optimize("O3")		
ios_base::sync_with_stdio(0);		

```
cin.tie(nullptr);
cout.tie(nullptr);
# Python
import sys
input = sys.stdin.readline
sys.stdout.write("-----")
```

2.2 SublimeBuild [13 lines] - 54e2df12

```
{
    "working_dir": "${file_path}",
    "selector": "source.c++, source.cpp",
    "shell": true,

    "windows": {
        "shell_cmd": "g++ -std=c++17 -O2 -Wall \"${file}\" -o\n\"${file_base_name}.exe\" \"&& \"${file_base_name}.exe\" <\nin.txt > out.txt 2> error.txt"
    },

    "linux": {
        "shell_cmd": "g++ -std=c++17 -O2 -Wall \"${file}\" -o\n\"${file_base_name}\" \"&& \"${file_base_name}\" <\nin.txt > out.txt 2> error.txt"
    }
}
```

2.3 VScode setup [7 lines] - 3e225a41

```
{
    "key": "f5",
    "command": "workbench.action.terminal.sendSequence",
    "args": {
        "text": "clear && g++ ${fileBasenameNoExtension}.cpp -o\n${fileBasenameNoExtension} &&\n./${fileBasenameNoExtension} < in.txt > out.txt\n"
    }
} // need "files.autoSaveDelay": 100
```

2.4 W_Debug [52 lines] - b6219639

```
void __print(int x) { cerr << x; }
void __print(long x) { cerr << x; }
void __print(long long x) { cerr << x; }
void __print(unsigned x) { cerr << x; }
void __print(unsigned long x) { cerr << x; }
void __print(unsigned long long x) { cerr << x; }
void __print(float x) { cerr << x; }
void __print(double x) { cerr << x; }
void __print(long double x) { cerr << x; }
void __print(char x) { cerr << '\\' << x << '\\'; }
void __print(const char *x) { cerr << '\"' << x << '\"'; }
void __print(const string &x) { cerr << '\"' << x << '\"'; }
void __print(bool x) { cerr << (x ? "true" : "false"); }
template <size_t N>
void __print(const bitset<N> &x) { cerr << x; }
template <typename T, typename V>
void __print(const pair<T, V> &x)
{
    cerr << '{';
    __print(x.first);
    cerr << ',';
    __print(x.second);
    cerr << '}';
}
template <typename T>
void __print(const T &x)
{
    int f = 0;
    cerr << '{';
    for (auto &i : x)
        cerr << (f++ ? ", " : ""), __print(i);
    cerr << "}";
}
void _print() { cerr << "]\n"; }
template <typename T, typename... V>
void _print(T t, V... v)
{
```

```
__print(t);
if (sizeof...(v))
    cerr << ", ";
__print(v...);
}
#ifdef ONLINE_JUDGE
#define debug(x...) \
    cerr << "[" << #x << "]" = ["; \n"; \
    __print(x)
#else
#define debug(x...)
#endif

// freopen("input.txt", "r", stdin);
// freopen("output.txt", "w", stdout);
```

2.5 stress_test [37 lines] - e1b7e560

```
//random number generator
#include <random>
#include <chrono>

mt19937 rng(chrono::steady_clock::
    now().time_since_epoch().count());
long long rnd(long long l, long long r) {
    return uniform_int_distribution<long long>(l, r)(rng);
}
```

// stress.sh file (chmod +x stress.sh)

```
#!/usr/bin/env bash
g++ -g "$1".cpp -DONPC -o "$1"
g++ -g "$2".cpp -DONPC -o "$2"
g++ -g "$3".cpp -DONPC -o "$3"
for ((testNum=0;testNum<$4;testNum++))
do
    ./$3 > stdinput
    ./$2 < stdinput > outSlow
    ./$1 < stdinput > outWrong
    H1=`md5sum outWrong`
    H2=`md5sum outSlow`
    if !(cmp -s "outWrong" "outSlow")
    then
        echo "Error found!"
        echo "Input:"
        cat stdinput
        echo "Wrong Output:"
        cat outWrong
        echo "Slow Output:"
        cat outSlow
        exit
    fi
done
echo Passed $4 tests
# ./stress.sh wrong correct gen times
```

2.6 vimrc [12 lines] - 1f47305c

```
"install xclip, vim-gtk3
set nosp ai bs=2 hls ic is lbr ls=2 mouse=a nu ru sc scs smd
so=3 sw=2 ts=2 rnu
filetype plugin indent on
syn on
map gA m'ggVG"+y''
inoremap {<CR> {<CR>}<Esc>ko
nnoremap = mzgg=G`z
vnoremap <C-k> :s/~/s*\zs/\\/ /<CR>:nohl<CR>
nnoremap <C-k> :s/~/s*\zs/\\/ /<CR>:nohl<CR>
vnoremap <C-l> :s/~/s*\zs/\\/ //<CR>:nohl<CR>
nnoremap <C-l> :s/~/s*\zs/\\/ //<CR>:nohl<CR>
autocmd FileType cpp map <F9> :w<CR> :!clear; g++ --std=c++17
% -DONPC -o %:r && ./%:r<CR>
```

3 Data Structure

3.1 DSU [31 lines] - ca62a11d

```
class DisjointSet {
    vector<int> par, sz, minElmt, maxElmt, cntElmt;
```

```
public:
DisjointSet(int n) {
    par.resize(n + 1);
    sz.resize(n + 1, 1);
    minElmt.resize(n + 1);
    maxElmt.resize(n + 1);
    cntElmt.resize(n + 1, 1);
    for (int i = 1; i <= n; i++) par[i] = minElmt[i] =
        maxElmt[i] = i;
}
int findUPar(int u) {
    if (u == par[u]) return u;
    return par[u] = findUPar(par[u]);
}
void unionBySize(int u, int v) {
    int pU = findUPar(u);
    int pV = findUPar(v);
    if (pU == pV) return;
    if (sz[pU] < sz[pV]) swap(pU, pV);
    par[pV] = pU;
    sz[pU] += sz[pV];
    cntElmt[pU] += cntElmt[pV];
    minElmt[pU] = min(minElmt[pU], minElmt[pV]);
    maxElmt[pU] = max(maxElmt[pU], maxElmt[pV]);
}
int getMinElementIntheSet(int u) { return
    minElmt[findUPar(u)]; }
int getMaxElementIntheSet(int u) { return
    maxElmt[findUPar(u)]; }
int getNumofElementIntheSet(int u) { return
    cntElmt[findUPar(u)]; }
};
```

3.2 Fenwick Tree [38 lines] - f052149b

```
int fenwick[N];
void update(int ind, int val) {
    while (ind < N) {
        fenwick[ind] += val;
        ind += ind & -ind;
    }
}
int query(int ind) {
    int sum = 0;
    while (ind > 0) {
        sum += fenwick[ind];
        ind -= ind & -ind;
    }
    return sum;
}

struct Fenwick2D {
    int n, m;
    vector<vector<int>> t;
    Fenwick2D(int n, int m) : n(n), m(m), t(n + 1, vector<int>(m
        + 1, 0)) {}

    void update(int i, int j, int v) {
        for (int x = i; x <= n; x += x & -x)
            for (int y = j; y <= m; y += y & -y) t[x][y] += v;
    }

    int query(int i, int j) {
        int s = 0;
        for (int x = i; x > 0; x -= x & -x)
            for (int y = j; y > 0; y -= y & -y) s += t[x][y];
        return s;
    }

    int query(int r1, int c1, int r2, int c2) {
        return query(r2, c2) - query(r1 - 1, c2) - query(r2, c1 -
            1) +
            query(r1 - 1, c1 - 1);
    }
};
```

3.3 MO with Update [61 lines] - f9eca766

```
// 1 indexed
// Complexity:  $O(S \times Q + Q \times \frac{N^2}{S^2})$ 
//  $S = (2*n^2)^{(1/3)}$ 
const int block_size = 2720; // 4310 for 2e5
const int mx = 1e5 + 5;
struct Query {
    int L, R, T, id;
    Query() {}
    Query(int _L, int _R, int _T, int _id) : L(_L), R(_R),
        T(_T), id(_id) {}
    bool operator<(const Query& x) const {
        if (L / block_size == x.L / block_size) {
            if (R / block_size == x.R / block_size) return T < x.T;
            return R / block_size < x.R / block_size;
        }
        return L / block_size < x.L / block_size;
    }
} Q[mx];
struct Update {
    int pos;
    int old, cur;
    Update() {}
    Update(int _p, int _o, int _c) : pos(_p), old(_o), cur(_c) {}
} U[mx];
int ans[mx];
inline void add(int id) {}
inline void remove(int id) {}
inline void update(int id, int L, int R) {
    // If update position is in current range [L,R]
    if (L <= U[id].pos && U[id].pos <= R) {
        remove(U[id].pos); // Remove old value
        arr[U[id].pos] = U[id].cur; // Apply update
        add(U[id].pos); // Add new value
    } else {
        arr[U[id].pos] = U[id].cur; // Just update array
    }
}
inline void undo(int id, int L, int R) {
    // Similar to update but revert to old value
    if (L <= U[id].pos && U[id].pos <= R) {
        remove(U[id].pos);
        arr[U[id].pos] = U[id].old;
        add(U[id].pos);
    } else {
        arr[U[id].pos] = U[id].old;
    }
}
inline int get() {}
void MO(int nq, int nu) {
    sort(Q + 1, Q + nq + 1);
    int L = 1, R = 0, T = nu;
    for (int i = 1; i <= nq; i++) {
        Query q = Q[i];
        while (T < q.T) update(++T, L, R);
        while (T > q.T) undo(T--, L, R);
        while (L > q.L) add(--L);
        while (R < q.R) add(++R);
        while (L < q.L) remove(L--);
        while (R > q.R) remove(R--);
        ans[q.id] = get();
    }
}
```

3.4 MO [58 lines] - 727b1684

```
const int N = 1e6 + 5;
int arr[N], freq[N], cnt = 0;
```

```
void remove(int idx) {
    int v = arr[idx];
    freq[v]--;
    if (freq[v] == 0) cnt--;
}
```

```
void add(int idx) {
    int v = arr[idx];
    freq[v]++;
    if (freq[v] == 1) cnt++;
}
int get_answer() {
    return cnt;
}

int block_size = 700;

struct Query {
    int l, r, idx;
    bool operator<(Query other) const {
        if (l / block_size != other.l / block_size) return l /
            block_size < other.l / block_size;
        return r < other.r;
    }
};

vector<int> mo_s_algorithm(vector<Query> queries) {
    vector<int> answers(queries.size());
    sort(queries.begin(), queries.end());

    // TODO: initialize data structure

    int cur_l = 0;
    int cur_r = -1;
    // invariant: data structure will always reflect the range
    [cur_l, cur_r]
    for (Query q : queries) {
        while (cur_l > q.l) {
            cur_l--;
            add(cur_l);
        }
        while (cur_r < q.r) {
            cur_r++;
            add(cur_r);
        }
        while (cur_l < q.l) {
            remove(cur_l);
            cur_l++;
        }
        while (cur_r > q.r) {
            remove(cur_r);
            cur_r--;
        }
        answers[q.idx] = get_answer();
    }
    return answers;
}
```

3.5 Persistent Segment Tree [114 lines] - 861d1673

```
#include <bits/stdc++.h>
using namespace std;
#define i64 long long
#ifdef ONPC
#include "mydebug.h"
#else
#define debug(...) 1
#endif
```

```
const int N = 30000 + 5;
int arr[N];
struct Node {
    int val;
    Node* lc;
    Node* rc;
    Node() {
        lc = NULL;
        rc = NULL;
    }
};
Node* versions[N];
```

```
Node* build(int l, int r) {
    Node* node = new Node();
    if (l == r) {
        node->val = 0;
        return node;
    }
    int mid = (l + r) / 2;
    node->lc = build(l, mid);
    node->rc = build(mid + 1, r);
    node->val = node->lc->val + node->rc->val;
    return node;
}
```

```
int query(Node* node, int l, int r, int ql, int qr) {
    if (l > qr || r < ql) return 0;
    if (ql <= l && r <= qr) {
        return node->val;
    }
    int mid = (l + r) / 2;
    int left = query(node->lc, l, mid, ql, qr);
    int right = query(node->rc, mid + 1, r, ql, qr);
    return left + right;
}
```

```
Node* update(Node* node, int l, int r, int ind, int val) {
    Node* newnode = new Node();
    if (l == r) {
        newnode->val = node->val + val;
        return newnode;
    }
    int mid = (l + r) / 2;
    if (l <= ind && ind <= mid) {
        newnode->lc = update(node->lc, l, mid, ind, val);
        newnode->rc = node->rc;
    } else {
        newnode->lc = node->lc;
        newnode->rc = update(node->rc, mid + 1, r, ind, val);
    }
    newnode->val = newnode->lc->val + newnode->rc->val;
    return newnode;
}
```

```
void solve() {
    int n;
    cin >> n;
    vector<int> comp;
    for (int i = 1; i <= n; i++) {
        cin >> arr[i];
        comp.push_back(arr[i]);
    }
```

```
sort(comp.begin(), comp.end());
comp.erase(unique(comp.begin(), comp.end()), comp.end());
int sz = comp.size();
```

```
for (int i = 1; i <= n; i++) {
    arr[i] = lower_bound(comp.begin(), comp.end(), arr[i]) -
        comp.begin();
}
```

```
versions[0] = build(0, sz - 1);
```

```
for (int v = 1; v <= n; v++) {
    versions[v] = update(versions[v - 1], 0, sz - 1, arr[v],
        +1);
}
```

```
int q;
cin >> q;
vector<int> ans(q + 1);
for (int i = 1; i <= q; i++) {
    int a, b, c;
    cin >> a >> b >> c;
```

```

int l = a ^ ans[i - 1];
int r = b ^ ans[i - 1];
int k = c ^ ans[i - 1];
l = max(l, 1);
r = min(n, r);
if (l > r) {
    ans[i] = 0;
    continue;
}
k = upper_bound(comp.begin(), comp.end(), k) -
    comp.begin();
if (k >= sz) {
    ans[i] = 0;
    continue;
}
ans[i] = query(versions[r], 0, sz - 1, k, sz - 1) -
    query(versions[l - 1], 0, sz - 1, k, sz - 1);
}
for (int i = 1; i <= q; i++) {
    cout << ans[i] << '\n';
}
}

```

3.6 SQRT Decomposition [69 lines] - a6182018

// <https://www.spoj.com/problems/GIVEAWAY/>
// 0 l r k : count of numbers >= k in [l,r]
// 1 idx val : update a[idx] = val

```

const int N = 5e5 + 5, BLK = 700;
vector<int> block[N / BLK + 5];
int get_block(int idx) { return idx / BLK; }

void solve() {
    int n;
    cin >> n;
    vector<int> a(n);
    for (int i = 0; i < n; i++) cin >> a[i];
    for (int i = 0; i < n; i++) {
        block[get_block(i)].push_back(a[i]);
    }
    int last_blk = get_block(n - 1);
    for (int i = 0; i <= last_blk; i++) {
        sort(block[i].begin(), block[i].end());
    }
    int q;
    cin >> q;
    while (q--) {
        int ty;
        cin >> ty;
        if (ty == 0) {
            // query
            // O(sqrt(n) log(sqrt(n)))
            int l, r, k;
            cin >> l >> r >> k;
            l--, r--;
            int bl = get_block(l);
            int br = get_block(r);
            int ans = 0;
            if (bl == br) {
                // just brute force
                for (int i = l; i <= r; i++) {
                    if (a[i] >= k) ans++;
                }
            } else {
                for (int i = l; get_block(i) < (bl + 1); i++) {
                    if (a[i] >= k) ans++;
                }
                for (int i = r; get_block(i) > (br - 1); i--) {
                    if (a[i] >= k) ans++;
                }
                for (int b = bl + 1; b < br; b++) {

```

```

// sorted vector, [1, 5, 10, 20], k = 6,
// ans += 2
auto it = lower_bound(block[b].begin(),
    block[b].end(), k);
ans += block[b].end() - it;
}
}
cout << ans << '\n';
} else {
    // update
    // O(sqrt(n) log(sqrt(n)))
    int idx, val;
    cin >> idx >> val;
    idx--;
    int old = a[idx];
    int b = get_block(idx);
    auto it = lower_bound(block[b].begin(),
        block[b].end(), old);
    *it = val;
    sort(block[b].begin(), block[b].end());
    a[idx] = val;
}
}
}
}

```

3.7 Segment Tree Lazy [73 lines] - 81e4cf40

```

struct SegTree {
    int n;
    vector<ll> tree, lazy;
    const ll NEUTRAL = 0;

    SegTree(int n) : n(n), tree(4 * n, NEUTRAL), lazy(4 * n, 0) {}

    ll combine(ll left, ll right) {
        return left + right;
    }

    void push(int v, int tl, int tr) {
        if (lazy[v] == 0) return;
        int m = (tl + tr) / 2;
        tree[2 * v] += lazy[v] * (m - tl + 1);
        lazy[2 * v] += lazy[v];
        tree[2 * v + 1] += lazy[v] * (tr - m);
        lazy[2 * v + 1] += lazy[v];
        lazy[v] = 0;
    }

    void build(const vector<int>& a, int v = 1, int tl = 0, int
        tr = -1) {
        if (tr < 0) tr = n - 1;
        if (tl == tr) {
            tree[v] = a[tl];
            return;
        }
        int m = (tl + tr) / 2;
        build(a, 2 * v, tl, m);
        build(a, 2 * v + 1, m + 1, tr);
        tree[v] = combine(tree[2 * v], tree[2 * v + 1]);
    }

    void update(int l, int r, int val, int v = 1, int tl = 0,
        int tr = -1) {
        if (tr < 0) tr = n - 1;
        if (l > tr || r < tl) return;
        if (l <= tl && tr <= r) {
            tree[v] += 1LL * val * (tr - tl + 1);
            lazy[v] += val;
            return;
        }
        push(v, tl, tr);
        int m = (tl + tr) / 2;
        update(l, r, val, 2 * v, tl, m);
        update(l, r, val, 2 * v + 1, m + 1, tr);
    }

```

```

    tree[v] = combine(tree[2 * v], tree[2 * v + 1]);
}

ll query(int l, int r, int v = 1, int tl = 0, int tr = -1) {
    if (tr < 0) tr = n - 1;
    if (l > tr || r < tl) return NEUTRAL;
    if (l <= tl && tr <= r) return tree[v];
    push(v, tl, tr);
    int m = (tl + tr) / 2;
    return combine(query(l, r, 2 * v, tl, m), query(l, r, 2 *
        v + 1, m + 1, tr));
}

void pointSet(int pos, int val, int v = 1, int tl = 0, int
    tr = -1) {
    if (tr < 0) tr = n - 1;
    if (tl == tr) {
        tree[v] = val;
        lazy[v] = 0;
        return;
    }
    push(v, tl, tr);
    int m = (tl + tr) / 2;
    if (pos <= m)
        pointSet(pos, val, 2 * v, tl, m);
    else
        pointSet(pos, val, 2 * v + 1, m + 1, tr);
    tree[v] = combine(tree[2 * v], tree[2 * v + 1]);
}
};

```

3.8 Segment Tree [53 lines] - 75840d89

```

constexpr int N = 100005;
i64 arr[N], seg[4 * N];

i64 combine(i64 l, i64 r) { return l + r; }
void build(int node, int low, int high) {
    if (low == high) {
        seg[node] = arr[low];
        return;
    }
    int mid = (low + high) / 2;
    build(2 * node, low, mid);
    build(2 * node + 1, mid + 1, high);
    seg[node] = combine(seg[2 * node], seg[2 * node + 1]);
}

i64 query(int node, int low, int high, int l, int r) {
    if (low >= l && high <= r) return seg[node];
    if (low > r || high < l) return 0; // check it!
    int mid = (low + high) / 2;
    i64 left = query(2 * node, low, mid, l, r);
    i64 right = query(2 * node + 1, mid + 1, high, l, r);
    return combine(left, right);
}

void update(int node, int low, int high, int pos, int val) {
    if (low == high) {
        seg[node] = val;
        return;
    }
    int mid = (low + high) / 2;
    if (low <= pos && pos <= mid)
        update(2 * node, low, mid, pos, val);
    else
        update(2 * node + 1, mid + 1, high, pos, val);
    seg[node] = combine(seg[2 * node], seg[2 * node + 1]);
}

// Maximum subarray sum in range [l, r]
struct Node {
    i64 sum, pref, suff, ans;
    Node() {
        sum = 0;
        pref = suff = ans = -inf;
    }
}

```

```
};
Node seg[4 * N];
Node combine(Node l, Node r) {
    Node res;
    res.sum = l.sum + r.sum;
    res.pref = max(l.pref, l.sum + r.pref);
    res.suff = max(r.suff, r.sum + l.suff);
    res.ans = max({l.ans, r.ans, res.pref, res.suff, res.sum});
    res.ans = max(res.ans, l.suff + r.pref);
    return res;
}
```

3.9 Special Range Query Example Fixing Right Position

[84 lines] - 5a580ece

```
/*
https://codeforces.com/gym/106107/problem/B
We need to count "good subarrays" [l, r] where:
1. Length >= 2
2. a[l] exists somewhere in b[l...r]
3. a[r] does NOT exist in b[l...r]
*/
```

```
#include <bits/stdc++.h>
using namespace std;
#define i64 long long
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;
```

```
const int N = 5e5 + 5;
i64 fenwick[N];
```

```
void update(int ind, int val) {
    while (ind < N) {
        fenwick[ind] += val;
        ind += ind & -ind;
    }
}
```

```
i64 query(int ind) {
    i64 sum = 0;
    while (ind > 0) {
        sum += fenwick[ind];
        ind -= ind & -ind;
    }
    return sum;
}
```

```
void solve() {
    int n;
    cin >> n;
    vector<int> comp;
    vector<int> a(n + 1), b(n + 1);
    gp_hash_table<int, vector<int>> mp;
    for (int i = 1; i <= n; i++) cin >> a[i];
    for (int i = 1; i <= n; i++) {
        cin >> b[i];
        mp[b[i]].push_back(i);
    }
    vector<int> rend(n + 1, n + 1);
    for (int i = 1; i <= n; i++) {
        if (mp[a[i]].empty()) continue;
        auto it = lower_bound(mp[a[i]].begin(), mp[a[i]].end(),
                               i);
        if (it == mp[a[i]].end()) continue;
        rend[i] = max(i + 1, *it); // length >= 2, l < r
    }
    vector<vector<int>> active(n + 2);
    for (int i = 1; i <= n; i++) {
        if (rend[i] <= n) {
            active[rend[i]].push_back(i); // i to rend[i] is a
            segment
        }
    }
    gp_hash_table<int, int> last;
    i64 seg_count = 0, ans = 0;
    for (int r = 1; r <= n; r++) {
```

```
        last[b[r]] = r;
        for (auto l : active[r]) {
            update(l, 1);
            seg_count++;
        }
        int last_pos = (last.find(a[r]) != last.end()) ?
            last[a[r]] : 0;
        i64 bad = query(last_pos);
        ans += seg_count - bad;
    }
    cout << ans << '\n';
}
```

```
// clear fenwick tree
for (int r = 1; r <= n; r++) {
    for (auto l : active[r]) update(l, -1);
}
}

int32_t main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);
    int t = 1;
    cin >> t;
    for (int i = 1; i <= t; i++) {
        // cout << "Case " << i << ": ";
        solve();
    }
}
```

3.10 Sqrt Tricks [8 lines] - 8f8b610b

- Size of the block is not always Sqrt, adjust it as necessary. if $O(n/b+b)$ then take $n/b = b$ and calculate b .
- MO's Algorithm
 - *it is possible to solve a Mo problem without any remove operation. For L in one block R only increases, for every range we can start L from the last of that block
- Sqrt Decomposition by time of queries.
 - *keep overall solution and sqrt(n) updates in a vector and for a query iterate over all of them, when the vector size exceeds sqrt(n) you can add these updates with overall solution using $O(n)$
- If sum of N positive numbers are S, there are at most sqrt(S) distinct values.
- Randomization
- Baby step, gaint step

3.11 Trie Bit [53 lines] - 0faa3353

```
const int BT = 32;
class Node {
public:
    Node* child[2];
    int cnt;
    Node() {
        cnt = 0;
        for (int i = 0; i < 2; i++) child[i] = NULL;
    }
};
Node *root = new Node();

void insert(Node* node, int x) {
    for (int i = BT - 1; i >= 0; i--) {
        int r = (x >> i) & 1LL;
        if (node->child[r] == NULL) node->child[r] = new Node();
        node = node->child[r];
        node->cnt++;
    }
} // insert(root, x);

int query(Node* node, int x) {
    int mx = 0;
    for (int i = BT - 1; i >= 0; i--) {
        int r = (x >> i) & 1LL;
        if (node->child[r ^ 1] == NULL || node->child[r ^ 1]->cnt
            == 0) {
            if (node->child[r] == NULL) return mx;
            node = node->child[r];
        }
    }
}
```

```
    } else {
        mx |= (1 << i);
        node = node->child[r ^ 1];
    }
}
return mx;
} // query(root, x), get max xor

void remove(Node* node, int x) {
    for (int i = BT - 1; i >= 0; i--) {
        int r = (x >> i) & 1LL;
        if (node->child[r] == NULL) return;
        node = node->child[r];
        node->cnt--;
    }
} // remove(root, x)

void clearTrie(Node* node) {
    if (node == nullptr) return;
    clearTrie(node->child[0]);
    clearTrie(node->child[1]);
    delete node;
}
// clearTrie(root), delete full trie
// root = new Node()
```

4 Dynamic Programming

4.1 All Possible Subarray Sum [6 lines] - 18ee4984

```
bitset<100005> bs = 1;
for (auto i : a) {
    bs |= (bs << i); // if previous 1 value pos is possible now
                    // ith bit or ith sm
                    // is also possible
}
cout << bs.count() - 1 << endl;
```

4.2 CHT [48 lines] - bc8514ba

```
#define x first
#define y second
// Warning: replace __int128 with long double if using floating
// point lines or if unsure about the compiler
// Queries maximum by default, pass false in constructor to
// query minimum
// Line = pair (m, c) for a line y = mx + c
// Make sure lines are inserted in non-decreasing order for
// maximum queries, and non-increasing order for minimum
// queries
template<typename T, class Line = pair<T, T>> // Make sure
multiplications fit in T
class CHT {
private:
    vector<Line> lines;
    bool maximum;
public:
    CHT(bool maximum = true) { this->maximum = maximum; }
    void addLine(Line line) {
        if (!maximum) line.first = -line.first, line.second =
            -line.second;
        insert(line);
    }
    T query(const T &x) {
        assert(!lines.empty());

        int L = 0, R = lines.size() - 1;
        while (L != R) {
            int mid1 = L + (R - L) / 3;
            int mid2 = R - (R - L) / 3;
            if (lines[mid1].x * x + lines[mid1].y
                >= lines[mid2].x * x + lines[mid2].y
            ) R = mid2 - 1;
            else L = mid1 + 1;
        }
    }
}
```

```

    T res = lines[L].x * x + lines[L].y;
    return maximum ? res : -res;
}
private:
bool bad(const Line &line1, const Line &line2, const Line
&line3) {
    return __int128(line3.y - line1.y) * __int128(line1.x -
line2.x)
        <= __int128(line2.y - line1.y) * __int128(line1.x
- line3.x);
}
void insert(const Line &line) {
    while (lines.size() > 0 && lines.back().x == line.x)
        lines.pop_back();
    lines.push_back(line);
    int sz = lines.size();
    while (sz >= 3 && bad(lines[sz - 3], lines[sz - 2],
lines[sz - 1])) {
        lines.erase(lines.end() - 2);
        sz--;
    }
};

```

4.3 Digit DP [16 lines] - 20d824ca

// Counts the values in $[L, R]$ with some property: $f(R) - f(L - 1)$.

// State: position, still hugging the upper bound, seen a non-zero digit yet.

string s; // the upper bound, as a digit string

int dp[20][2][2];

```

int f(int pos, int tight, int started) {
    if (pos == (int)s.size()) return started; // <- put the
        property here
    int& ans = dp[pos][tight][started];
    if (ans != -1) return ans;
    ans = 0;
    int hi = tight ? s[pos] - '0' : 9;
    for (int d = 0; d <= hi; d++)
        ans += f(pos + 1, tight && d == hi, started || d);
    return ans;
}
// memset(dp, -1, sizeof dp); s = to_string(R); f(0, 1, 0);

```

4.4 Divide and Conquer DP [26 lines] - 1949a056

```

ll G,L;///total group,cell size
ll dp[8001][801],cum[8001];
ll C[8001];///value of each cell
inline ll cost(ll l,ll r){
    return(cum[r]-cum[l-1])*(r-l+1);
}
void fn(ll g,ll st,ll ed,ll r1,ll r2){
    if(st>ed)return;
    ll mid=(st+ed)/2,pos=-1;
    dp[mid][g]=inf;
    for(int i=r1;i<=r2;i++){
        ll tcost=cost(i,mid)+dp[i-1][g-1];
        if(tcost<dp[mid][g]){
            dp[mid][g]=tcost,pos=i;
        }
    }
    fn(g,st,mid-1,r1,pos);
    fn(g,mid+1,ed,pos,r2);
}
int main(){
    for(int i=1;i<=L;i++)
        cum[i]=cum[i-1]+C[i];
    for(int i=1;i<=L;i++)
        dp[i][1]=cost(1,i);
    for(int i=2;i<=G;i++)fn(i,1,L,1,L);
}

```

4.5 LCIS $o(n*m)$ [22 lines] - 0a55ae12

```
int a[100] = {0}, b[100] = {0}, f[100] = {0};
```

```

int n = 0, m = 0;
int main(void) {
    cin >> n;
    for (int i = 1; i <= n; i++) cin >> a[i];
    cin >> m;
    for (int i = 1; i <= m; i++) cin >> b[i];
    for (int i = 1; i <= n; i++) {
        int k = 0;
        for (int j = 1; j <= m; j++) {
            if (a[i] > b[j] && f[j] > k)
                k = f[j];
            else if (a[i] == b[j] && k + 1 > f[j])
                f[j] = k + 1;
        }
    }
    int ans = 0;
    for (int i = 1; i <= m; i++)
        if (f[i] > ans) ans = f[i];
    cout << ans << endl;
    return 0;
}

```

4.6 LCS [62 lines] - 2e950914

```

// LCS DP Table and LCS Length
vector<vector<int>>> dp(s.size() + 1, vector<int>(t.size() +
1));
for (int i = 1; i <= s.size(); i++) {
    for (int j = 1; j <= t.size(); j++) {
        if (s[i - 1] == t[j - 1])
            dp[i][j] = dp[i - 1][j - 1] + 1;
        else
            dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
    }
}
cout << dp[s.size()][t.size()] << "\n";

// Any LCS String
string a;
int i = s.size();
int j = t.size();
while (i > 0 && j > 0) {
    if (s[i - 1] == t[j - 1]) {
        a += s[i - 1];
        i--, j--;
    } else if (dp[i][j - 1] > dp[i - 1][j])
        j--;
    else
        i--;
}
reverse(a.begin(), a.end());
cout << a << "\n";

```

```

// Lexicographically smallest LCS
vector<vector<string>>> dpp(s.size() + 1,
vector<string>(t.size() + 1));
for (int i = 1; i <= s.size(); i++) {
    for (int j = 1; j <= t.size(); j++) {
        if (s[i - 1] == t[j - 1])
            dpp[i][j] = dpp[i - 1][j - 1] + s[i - 1];
        else if (dp[i][j - 1] < dp[i - 1][j])
            dpp[i][j] = dpp[i - 1][j];
        else if (dp[i][j - 1] > dp[i - 1][j])
            dpp[i][j] = dpp[i][j - 1];
        else
            dpp[i][j] = min(dpp[i][j - 1], dpp[i - 1][j]);
    }
}
cout << dpp[s.size()][t.size()] << "\n";

```

```

// number of distinct LCS sequences
int MOD = 1e9 + 7;
vector<vector<int>>> cnt(s.size() + 1, vector<int>(t.size() +
1, 1));
for (int i = 1; i <= s.size(); i++) {
    for (int j = 1; j <= t.size(); j++) {

```

```

        if (s[i - 1] == t[j - 1])
            cnt[i][j] = cnt[i - 1][j - 1];
        else if (dp[i][j - 1] < dp[i - 1][j])
            cnt[i][j] = cnt[i - 1][j];
        else if (dp[i][j - 1] > dp[i - 1][j])
            cnt[i][j] = cnt[i][j - 1];
        else {
            cnt[i][j] = cnt[i - 1][j] + cnt[i][j - 1];
            if (dp[i][j] == dp[i - 1][j - 1]) cnt[i][j] -= cnt[i -
1][j - 1];
            cnt[i][j] = (cnt[i][j] + MOD) % MOD;
        }
    }
}

```

4.7 LIS $O(n \log n)$ with full path [50 lines] - d910146b

```

vector<int> v = {7, 3, 5, 3, 6, 2, 9, 8};
vector<int> seq;
for (auto i : v) {
    auto id = lower_bound(seq.begin(), seq.end(), i);
    if (id == seq.end())
        seq.push_back(i);
    else
        seq[id - seq.begin()] = i;
}
cout << seq.size() << endl;

```

```

void lis(vector<int> &v) {
    int n = v.size();
    vector<int> dp(n + 1, 1), hash(n);
    int mx = 1, lastInd = 0;
    for (int i = 0; i < n; i++) {
        hash[i] = i;
        for (int prev = 0; prev < i; prev++) {
            if (v[i] > v[prev] && 1 + dp[prev] > dp[i]) {
                dp[i] = 1 + dp[prev];
                hash[i] = prev;
            }
        }
        if (mx < dp[i]) {
            mx = dp[i];
            lastInd = i;
        }
    }
    vector<int> printSeq;
    printSeq.push_back(v[lastInd]);
    while (hash[lastInd] != lastInd) {
        lastInd = hash[lastInd];
        printSeq.push_back(v[lastInd]);
    }
    reverse(printSeq.begin(), printSeq.end());
    cout << mx << "\n";
    for (int i : printSeq) cout << i << " ";
    cout << "\n";
}

```

```

// nlogn. segment tree max
vector<int> dp(n + 1);

for (int i = 0; i < n; i++) {
    dp[i] = 1;
    int mx = query(0, 1, N, 1, a[i] - 1);
    dp[i] = max(dp[i], mx + 1);
    update(0, 1, N, a[i], dp[i]);
}
int ans = *max_element(dp.begin(), dp.end());
}

```

4.8 MCM [26 lines] - e8d610dd

```

const int N = 1005;
vector<int> v;
int dp[N][N], mark[N][N];
int MCM(int i, int j) {
    if (i == j) return dp[i][j] = 0;
    if (dp[i][j] != -1) return dp[i][j];

```

```
int mn = INT_MAX;
for (int k = i; k < j; k++) {
    int x = mn;
    mn = min(mn, MCM(i, k) + MCM(k + 1, j) + v[i - 1] * v[k]
        * v[j]);
    if (x != mn) mark[i][j] = k;
}
return dp[i][j] = mn;
}
void print_order(int i, int j) {
    if (i == j)
        cout << "X" << i;
    else {
        cout << "(";
        print_order(i, mark[i][j]);
        print_order(mark[i][j] + 1, j);
        cout << ")";
    }
}
// memset(dp, -1, sizeof dp);
// print_order(i, n);
```

4.9 Maximum Submatrix [34 lines] - 30548291

```
int a[150][150] = {0};
int c[200] = {0};
int maxarray(int n) {
    int b = 0, sum = -100000000;
    for (int i = 1; i <= n; i++) {
        if (b > 0)
            b += c[i];
        else
            b = c[i];
        if (b > sum) sum = b;
    }
    return sum;
}
int maxmatrix(int n) {
    int sum = -100000000, max = 0;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) c[j] = 0;
        for (int j = i; j <= n; j++) {
            for (int k = 1; k <= n; k++) c[k] += a[j][k];
            max = maxarray(n);
            if (max > sum) sum = max;
        }
    }
    return sum;
}
int main(void) {
    int n = 0;
    cin >> n;
    for (int i = 1; i <= n; i++)
        for (int j = 1; j <= n; j++) cin >> a[i][j];
    cout << maxmatrix(n);
    return 0;
}
```

4.10 SOS DP [63 lines] - 2e55972d

```
// sum over subsets
// fact: need sum of submasks
void SOS(vector<int>& v) {
    const int BITS = log2(*max_element(v.begin(), v.end())) + 1;
    vector<int> freq(1 << BITS, 0);
    for (int mask : v) freq[mask]++;

    vector<vector<int>> dp(BITS + 1, vector<int>(1 << BITS, 0));
    for (int mask = 0; mask < 1 << BITS; mask++) {
        dp[0][mask] = freq[mask];

        for (int bits = 1; bits <= BITS; bits++) {
            for (int mask = 0; mask < 1 << BITS; mask++) {
                if ((mask & (1 << (bits - 1))) == 0) {
                    dp[bits][mask] = dp[bits - 1][mask];
                }
            }
        }
    }
}
```

```
} else {
    int other_mask = mask - (1 << (bits - 1));
    dp[bits][mask] = dp[bits - 1][mask] + dp[bits - 1][other_mask];
}
}
}

for (int mask : v) cout << dp[BITS][mask] << '\n';
// O(BITS * 2 ^ BITS)
// dp[bits][mask] means left most 'bits' of submasks can be
// differ (submask)
// for dp[1][1] 01, 00 are not allow because leftmost 1 bit
// can be differ. 10
// and 11 are allowed. for travarsing all the submask of a mask
// we can use
// submask = mask
// do {
//     submask = (submask - 1) & mask
// } while (submask)
// O(3 ^ BITS)

const int MAXN = 1 << 20, MLOG = 20;
int dp[MAXN], freq[MAXN];

// Sum over subsets (SOS Down)
void forward1() {
    for (int bit = 0; bit < MLOG; bit++)
        for (int i = 0; i < MAXN; i++)
            if (i & (1 << bit)) dp[i] += dp[i ^ (1 << bit)];
}

void backward1() {
    for (int bit = 0; bit < MLOG; bit++)
        for (int i = MAXN - 1; i >= 0; i--)
            if (i & (1 << bit)) dp[i] -= dp[i ^ (1 << bit)];
}

// Sum over supersets (SOS Up)
void forward2() {
    for (int bit = 0; bit < MLOG; bit++)
        for (int i = MAXN - 1; i >= 0; i--)
            if (i & (1 << bit)) dp[i ^ (1 << bit)] += dp[i];
}

void backward2() {
    for (int bit = 0; bit < MLOG; bit++)
        for (int i = 0; i < MAXN; i++)
            if (i & (1 << bit)) dp[i ^ (1 << bit)] -= dp[i];
}
```

5 Game Theory

5.1 Points to be noted [14 lines] - 85ddc883

>[First Write a Brute Force solution]
 >Nim = all xor
 >Misere Nim = Nim + corner case: if all piles are 1, reverse(nim)
 >Bogus Nim = Nim
 >Staircase Nim = Odd indexed pile Nim (Even indexed pile doesn't matter, as one player can give bogus moves to drop all even piles to ground)
 >Sprague Grundy: [Every impartial game under the normal play convention is equivalent to a one-heap game of nim]
 Every tree = one nim pile = tree root value; tree leaf value = 0; tree node value = mex of all child nodes.
 [Careful: one tree node can become multiple new tree roots (multiple elements in one node), then the value of that node = xor of all those root values]
 >Hackenbush (Given a rooted tree; cut an edge in one move; subtree under that edge gets removed; last player to cut wins):
 Colon: $//G(u) = (G(v_1) + 1) \oplus (G(v_2) + 1) \oplus \dots [v_1, v_2, \dots \text{ are childs of } u]$

For multiple trees ans is their xor
 >Hackenbush on graph (instead of tree given an rooted graph):
 fusion: All edges in a cycle can be fused to get a tree structure; build a super node, connect some single nodes with that super node, number of single nodes is the number of edges in the cycle.
 Sol: [Bridge component tree] mark all bridges, a group of edges that are not bridges, becomes one component and contributes number of edges to the hackenbush. (even number of edges contributes 0, odd number of edges contributes 1)

6 Geometry

6.1 2D Convex Hull [67 lines] - 0b5204d1

```
struct point {
    ll x, y;
    bool operator==(point const& t) const { return (x == t.x && y == t.y); }
};

int orientation(point a, point b, point c) {
    ll d = (c.y - b.y) * (b.x - a.x) - (c.x - b.x) * (b.y - a.y);
    if (d < 0)
        return -1; // clockwise
    else if (d > 0)
        return 1; // anticlockwise
    return 0; // collinear
}

bool clockwise(point a, point b, point c, bool include_collinear) {
    include_collinear) {
        int o = orientation(a, b, c);
        if (o < 0) return true;
        if (o > 0) return false;
        return include_collinear;
    }
}

bool collinear(point a, point b, point c) { return orientation(a, b, c) == 0; }
bool cmp_points(point& p1, point& p2) {
    return make_pair(p1.x, p1.y) < make_pair(p2.x, p2.y);
}

ll distance_sq(point a, point b) {
    ll d = (a.x - b.x) * (a.x - b.x) + (a.y - b.y) * (a.y - b.y);
    return d;
}

void print(vector<point>& v) {
    for (auto it : v) {
        cout << "(" << it.x << ", " << it.y << ")" << "\n";
    }
}

vector<point> convex_hull(vector<point>& points, bool include_collinear = false) {
    int sz = (int)points.size();
    point p0 = *min_element(points.begin(), points.end(), cmp_points);
    vector<point> convex_hull_vector = points;
    sort(convex_hull_vector.begin(), convex_hull_vector.end(), [&p0](const point& p1, const point& p2) {
        int o = orientation(p0, p1, p2);
        if (o == 0) {
            return distance_sq(p0, p1) < distance_sq(p0, p2);
        }
        return o < 0;
    });
    if (include_collinear) {
        int idx = sz - 1;
        while (idx >= 0 && collinear(p0, convex_hull_vector[idx], convex_hull_vector.back()))
            idx--;
    }
}
```

```

    reverse(convex_hull_vector.begin() + idx + 1,
           convex_hull_vector.end());
}
vector<point> st;
for (int i = 0; i < sz; i++) {
    while (st.size() >= 2 &&
           !clockwise(st[st.size() - 2], st.back(),
                     convex_hull_vector[i],
                     include_collinear)) {
        st.pop_back();
    }
    st.push_back(convex_hull_vector[i]);
}
if (!include_collinear && st.size() == 2 && st[0] == st[1])
    st.pop_back();
}
return st;
}

```

6.2 Computational Geometry [74 lines] - a0efcafa

```

typedef long long int T;
typedef double Tf;
Tf eps = 1e-10;

// int sgn(Tf x)
// {
//     if (fabs(x) < eps) return 0;
//     if (x > 0) return 1;
//     return -1;
// }

int sgn(T x) {
    if (x == 0) return 0;
    if (x > 0)
        return 1;
    else
        return -1;
}

struct point {
    T x, y;
    point(T x = 0, T y = 0) {
        this->x = x;
        this->y = y;
    }
    point operator+(point p) { return point(x + p.x, y + p.y); }
    point operator-(point p) { return point(x - p.x, y - p.y); }
    Tf len() { return sqrt(x * x + y * y); }
    T norm() { return x * x + y * y; }
    point operator*(T a) { return point(x * a, y * a); }
    point operator/(T a) { return point(x / a, y / a); }
    // integer
    bool operator==(point p) { return p.x == x && p.y == y; }
    // float
    // bool operator==(point p) {
    //     return sgn(p.x - x) == 0 && sgn(p.y - y) == 0;
    // }
    bool operator<(point p) {
        if (x == p.x) return y < p.y;
        return x < p.x;
    }
    // float
    // bool operator<(point p) {
    //     if (sgn(p.x - x) == 0) return sgn(y - p.y) == -1;
    //     return sgn(x - p.x) == -1;
    // }
};
T dot(point a, point b) { return a.x * b.x + a.y * b.y; }
T cross(point a, point b) { return a.x * b.y - b.x * a.y; }
T orien(point a, point b, point c) { return cross(a - b, a - c); }

Tf area(point a, point b, point c) { return abs(orien(a, b, c)) / (Tf)2; }

```

```

struct segment {
    point a, b;
    bool onSegment(point p) {
        if (sgn(orien(a, b, p)) != 0) return false;
        return sgn(dot(p - a, p - b)) <= 0;
    }
    bool intersection(segment s) {
        if (onSegment(s.a)) return true;
        if (onSegment(s.b)) return true;
        if (s.onSegment(a)) return true;
        if (s.onSegment(b)) return true;
        int s1 = sgn(orien(a, b, s.a));
        int s2 = sgn(orien(a, b, s.b));
        int s3 = sgn(orien(s.a, s.b, a));
        int s4 = sgn(orien(s.a, s.b, b));
        if (s1 * s2 < 0 && s3 * s4 < 0) return true;
        return false;
    }
};
ostream& operator<<(ostream& os, point p) {
    return os << "(" << p.x << "," << p.y << ")";
}

```

7 Graph

7.1 Articulation Bridges [41 lines] - b120c1d4

```

const int N = 2e5 + 5;
vector < pair < int, int >> g[N];
int vis[N], st[N], low[N];
int clk, n;
vector < pair < int, int >> bridges;

void dfs(int u, int p, int edg) {
    vis[u] = 1;
    ++clk;
    st[u] = clk;
    low[u] = clk;
    for (auto [v, id]: g[u]) {
        // handled parallel edges between u, v
        if (v == p && id == edg) continue;
        if (vis[v]) {
            low[u] = min(low[u], st[v]);
        } else {
            dfs(v, u, id);
            low[u] = min(low[u], low[v]);
            if (st[u] < low[v]) {
                // edge u, v has bridge
                bridges.push_back({
                    u,
                    v
                });
            }
        }
    }
}

void find_bridges() {
    clk = 0;
    for (int i = 1; i <= n; i++) {
        vis[i] = 0;
        st[i] = low[i] = -1;
    }
    for (int i = 1; i <= n; i++) {
        if (!vis[i]) {
            dfs(i, 0, -1);
        }
    }
}

```

7.2 Articulation Points [40 lines] - 94a1ef65

```

const int N = 2e5 + 5;
vector < int > g[N];
int vis[N], st[N], low[N], adjComp[N];
int clk, n;

void dfs(int u, int p) {

```

```

    vis[u] = 1;
    ++clk;
    st[u] = clk;
    low[u] = clk;
    int child = 0;
    for (int v: g[u]) {
        if (v == p) continue;
        if (vis[v]) {
            low[u] = min(low[u], st[v]);
        } else {
            dfs(v, u);
            low[u] = min(low[u], low[v]);
            if (low[v] >= st[u] && p != 0) {
                adjComp[u]++;
            }
            child++;
        }
    }
    if (p == 0 && child > 1) {
        adjComp[u] = child - 1;
    }
}

void find_vertices() {
    clk = 0;
    for (int i = 1; i <= n; i++) {
        vis[i] = 0;
        st[i] = low[i] = -1;
    }
    for (int i = 1; i <= n; i++) {
        if (!vis[i]) {
            dfs(i, 0);
        }
    }
}

```

7.3 Bellmanford [22 lines] - b8e770fd

```

void bellmanFord(int num_of_nd, int src) {
    dist[src] = 0;
    for (int step = 0; step < num_of_nd; step) {
        for (int i = 1; i <= num_of_nd; i++) {
            for (auto it : adj[i]) {
                int u = i;
                int v = it.first;
                int wt = it.second;
                if (dist[u] != inf && ((dist[u] + wt) < dist[v])) {
                    if (step == num_of_nd - 1) {
                        cout << "Negative cycle found\n ";
                        return;
                    }
                    dist[v] = dist[u] + wt;
                    parent[v] = u;
                }
            }
        }
    }
    for (int i = 1; i <= num_of_nd; i++) cout << dist[i] << " ";
    cout << endl;
}

```

7.4 Block Cut Tree [83 lines] - b31a5c97

```

const int N = 2e5 + 5;

vector<int> g[N], bct[2 * N];
int vis[N], low[N], disc[N], is_art[N];
stack<int> stk;
vector<int> block_of[N];
int clk, n, m, bct_nodes, root;

void dfs(int u, int p) {
    vis[u] = 1;
    disc[u] = low[u] = ++clk;
    stk.push(u);
    int child = 0;

```

```

for (int v : g[u]) {
    if (v == p) continue;
    if (!vis[v]) {
        child++;
        dfs(v, u);
        low[u] = min(low[u], low[v]);

        if ((p == -1 && child > 1) || (p != -1 && low[v] >=
            disc[u])) {
            is_art[u] = 1;
            vector<int> block;
            while (stk.top() != v) {
                block.push_back(stk.top());
                stk.pop();
            }
            block.push_back(stk.top());
            stk.pop();
            block.push_back(u);

            int block_id = ++bct_nodes;
            for (int x : block) {
                block_of[x].push_back(block_id);
                if (is_art[x]) {
                    bct[x].push_back(block_id);
                    bct[block_id].push_back(x);
                }
            }
        } else {
            low[u] = min(low[u], disc[v]);
        }
    }
}

void build_bct() {
    clk = 0;
    bct_nodes = n;
    for (int i = 1; i <= n; i++) {
        vis[i] = is_art[i] = 0;
        block_of[i].clear();
        bct[i].clear();
    }
    for (int i = n + 1; i <= 2 * n; i++) bct[i].clear();

    for (int i = 1; i <= n; i++) {
        if (!vis[i]) {
            dfs(i, -1);
            if (!stk.empty()) {
                int block_id = ++bct_nodes;
                while (!stk.empty()) {
                    int x = stk.top();
                    stk.pop();
                    block_of[x].push_back(block_id);
                    if (is_art[x]) {
                        bct[x].push_back(block_id);
                        bct[block_id].push_back(x);
                    }
                }
            }
        }
    }

    root = n + 1;
    for (int i = 1; i <= n; i++) {
        if (is_art[i]) {
            root = i;
            break;
        }
    }
}

```

7.5 Centroid Decomposition [30 lines] - 2ab608d6

```

const int N = 1e5 + 5;
vector<int> g[N];
int sz[N], dead[N], cenpar[N];

```

```

void dfs_sz(int u, int p) {
    sz[u] = 1;
    for (int v : g[u]) {
        if (v == p || dead[v]) continue;
        dfs_sz(v, u);
        sz[u] += sz[v];
    }
}

int find_centroid(int num, int u, int p) {
    for (int v : g[u]) {
        if (v != p && !dead[v] && 2 * sz[v] > num) {
            return find_centroid(num, v, u);
        }
    }
    return u;
}

void decompose(int u, int p) {
    dfs_sz(u, p);
    int cent = find_centroid(sz[u], u, p);
    dead[cent] = true;
    cenpar[cent] = p;
    for (int v : g[cent]) {
        if (!dead[v]) {
            decompose(v, cent);
        }
    }
}

```

7.6 Dijkstra [22 lines] - 3621521c

```

void Dijkstra(int start) {
    // vector<pair<int, int>> adj[N];
    priority_queue<pair<int, int>, vector<pair<int, int>>,
        greater<pair<int, int>>> pq;
    pq.push({0, start});
    while (!pq.empty()) {
        auto it = pq.top();
        pq.pop();
        int wt = it.first;
        int u = it.second;
        if (vis[u])
            continue;
        vis[u] = 1;
        for (pair<int, int> i : adj[u]) {
            int adjWt = i.second;
            int adjNode = i.first;
            if (dist[adjNode] > wt + adjWt) {
                dist[adjNode] = wt + adjWt;
                pq.push({dist[adjNode], adjNode});
            }
        }
    }
}

```

7.7 Eulerian Circuit [16 lines] - b13e8b7b

```

unordered_map<int, int> Start, End, Val;
unordered_map<int, pair<int, int>> Range;
int start = 0;
void dfs(int node) {
    visited[node] = true;
    Start[node] = start++;
    for (auto child : adj[node]) {
        if (!visited[child]) dfs(child);
    }
    End[node] = start - 1;
}
dfs(1);
vector<int> FlatArray(start + 5);
for (auto i : Start) {
    FlatArray[i.second] = Val[i.first];
    Range[i.first] = {i.second, End[i.first]};
}

```

7.8 Floyd Warshall [13 lines] - 7e17addb

```

const int inf = 1e9;
const int MAXN = 505;
int dist[MAXN][MAXN];

```

```

void floydWarshall(int n) {
    for (int k = 0; k < n; ++k) {
        for (int i = 0; i < n; ++i) {
            for (int j = 0; j < n; ++j) {
                if (dist[i][k] < inf && dist[k][j] < inf)
                    dist[i][j] = min(dist[i][j], dist[i][k] +
                        dist[k][j]);
            }
        }
    }
}

```

7.9 Heavy Light Decomposition [103 lines] - ac779cb8

```

vector<pair<int, int>> g[N];
int edge_to_ch[N];
int par[N], dep[N], sz[N];

void dfs(int u, int p = 0) {
    sz[u] = 1, par[u] = p;
    dep[u] = dep[p] + 1;
    int mx = 0;
    for (auto &[v, idx]: g[u]) {
        if (v == p) continue;
        edge_to_ch[idx] = v;
        dfs(v, u);
        sz[u] += sz[v];
        if (sz[v] > mx) {
            mx = sz[v];
            swap(v, g[u][0].first);
        }
    }
}

int T, head[N], st[N], en[N];

void dfs_hld(int u, int p = 0) {
    st[u] = ++T;
    head[u] = (p != 0 && g[p][0].first == u) ? head[p] : u;
    for (auto [v, idx]: g[u]) {
        if (v == p) continue;
        dfs_hld(v, u);
    }
    en[u] = T;
}

int lca(int a, int b) {
    for (; head[a] != head[b]; b = par[head[b]]) {
        if (dep[head[a]] > dep[head[b]]) swap(a, b);
    }
    if (dep[a] > dep[b]) swap(a, b);
    return a;
}

int n;
// process node u upto it's ancestor a
// if excl is true, a won't process
i64 chain_process(int a, int u, bool excl = false) {
    i64 ans = 0;
    for (; head[u] != head[a]; u = par[head[u]]) {
        ans = max(ans, query(1, 1, n, st[head[u]], st[u]));
    }
    ans = max(ans, query(1, 1, n, st[a] + excl, st[u]));
    return ans;
}

// process path from node a to b
// if excl is true, lca won't process
i64 path_process(int a, int b, bool excl = false) {
    i64 ans = 0;
    for (; head[a] != head[b]; b = par[head[b]]) {
        if (dep[head[a]] > dep[head[b]]) swap(a, b);
        ans = max(ans, query(1, 1, n, st[head[b]], st[b]));
    }
    if (dep[a] > dep[b]) swap(a, b);
    ans = max(ans, query(1, 1, n, st[a] + excl, st[b]));
    return ans;
}

```

```

// update path from node u to v
// if excl is true, lca won't update
void path_update(int a, int b, int val, bool excl = false) {
    for (; head[a] != head[b]; b = par[head[b]]) {
        if (dep[head[a]] > dep[head[b]]) swap(a, b);
        update(1, 1, n, st[head[b]], st[b], val);
    }
    if (dep[a] > dep[b]) swap(a, b);
    update(1, 1, n, st[a] + excl, st[b], val);
}
/*
    dfs(1);
    head[1] = 1;
    dfs_hld(1);
    build(1, 1, n);
*/
void solve() {
    cin >> n;
    int q;
    cin >> q;
    for (int i = 1; i < n; i++) {
        int u, v, c;
        cin >> u >> v >> c;
        g[u].push_back({v, i});
        g[v].push_back({u, i});
        edge_cost[i] = c;
    }
    dfs(1);
    head[1] = 1;
    dfs_hld(1);
    for (int i = 1; i < n; i++) {
        int u = edge_to_ch[i];
        arr[st[u]] = edge_cost[i];
    }
    build(1, 1, n);
    while (q--) {
        int u, v, val;
        cin >> u >> v >> val;
        path_update(u, v, val, true);
        cout << path_process(u, v, true) << '\n';
    }
}

```

7.10 K'th Shortest path [40 lines] - 5a50a704

```

int m,n,deg[MM],source,sink,K,val[MM][12];
struct edge{
    int v,w;
}adj[MM][500];
struct info{
    int v,w,k;
    bool operator<(const info &b)const{
        return w>b.w;
    }
};
priority_queue<info,vector<info>>Q;
void kthBestShortestPath(){
    int i,j;
    info u,v;
    for(i=0;i<n;i++)
        for(j=0;j<K;j++)val[i][j]=inf;
    u.v=source,u.k=0,u.w=0;
    Q.push(u);
    while(!Q.empty()){
        u=Q.top();
        Q.pop();
        for(i=0;i<deg[u.v];i++){
            v.v=adj[u.v][i].v;
            int cost=adj[u.v][i].w+u.w;
            for(v.k=u.k;v.k<K;v.k++){
                if(cost==inf)break;
                if(val[v.v][v.k]>cost){
                    swap(cost,val[v.v][v.k]);
                    v.w=val[v.v][v.k];
                    Q.push(v);
                    break;
                }
            }
        }
    }
}

```

```

    }
    }
    for(v.k++;v.k<K;v.k++){
        if(cost==inf)break;
        if(val[v.v][v.k]>cost)swap(cost, val[v.v][v.k]);
    }
}
}
}
}

```

7.11 Kruskal [13 lines] - 7c746046

```

vector<pair<int, pair<int, int>>> Kruskal(
    vector<pair<int, pair<int, int>>>& edges, int n) {
    sort(edges.begin(), edges.end());
    vector<pair<int, pair<int, int>>> ans;
    DisjointSet D(n);
    for (auto it : edges) {
        if (D.findUPar(it.second.first) !=
            D.findUPar(it.second.second)) {
            ans.push_back({it.first, {it.second.first,
                it.second.second}});
            D.unionBySize(it.second.first, it.second.second);
        }
    }
    return ans;
}

```

7.12 LCA using Segment Tree [89 lines] - fa80d47b

```

struct LCA {
    vector<int> height, euler, first, segtree, parent;
    vector<bool> visited;
    vector<vector<int>> jump;

    int n;

```

```

    LCA(vector<vector<int>> &adj, int root = 0) {
        n = adj.size();
        height.resize(n);
        first.resize(n);
        parent.resize(n);
        euler.reserve(n * 2);
        visited.assign(n, false);
        dfs(adj, root);
        int m = euler.size();
        segtree.resize(m * 4);
        build(1, 0, m - 1);
    }

```

```

    jump.resize(n, vector<int>(32, -1));

```

```

    for(int i=0;i<n;i++) {
        jump[i][0] = parent[i];
    }

```

```

    for(int j=1;j<20;j++) {
        for(int i=0;i<n;i++) {
            int mid = jump[i][j-1];
            if(mid != -1) jump[i][j] = jump[mid][j-1];
        }
    }
}

```

```

void dfs(vector<vector<int>> &adj, int node, int h = 0) {
    visited[node] = true;
    height[node] = h;
    first[node] = euler.size();
    euler.push_back(node);
    for (auto to : adj[node]) {
        if (!visited[to]) {
            parent[to] = node;
            dfs(adj, to, h + 1);
            euler.push_back(node);
        }
    }
}
}
}

```

```

void build(int node, int b, int e) {
    if (b == e) {
        segtree[node] = euler[b];
    } else {
        int mid = (b + e) / 2;
        build(node << 1, b, mid);
        build(node << 1 | 1, mid + 1, e);
        int l = segtree[node << 1], r = segtree[node << 1 | 1];
        segtree[node] = (height[l] < height[r]) ? l : r;
    }
}

```

```

int query(int node, int b, int e, int L, int R) {
    if (b > R || e < L)
        return -1;
    if (b >= L && e <= R)
        return segtree[node];
    int mid = (b + e) >> 1;

    int left = query(node << 1, b, mid, L, R);
    int right = query(node << 1 | 1, mid + 1, e, L, R);
    if (left == -1)
        return right;
    if (right == -1)
        return left;
    return height[left] < height[right] ? left : right;
}

```

```

int lca(int u, int v) {
    int left = first[u], right = first[v];
    if (left > right)
        swap(left, right);
    return query(1, 0, euler.size() - 1, left, right);
}

```

```

int kthParent(int u, int k) {
    for(int i = 20; i >= 0; i--) {
        if(k & (1LL << i)) u = jump[u][i];
    }
    return u;
}
};

```

7.13 LCA [40 lines] - e4c6dfdc

```

const int N = 1e5 + 5, LG = 17;
vector<int> g[N];
int depth[N], parent[N][LG];
void dfs(int u, int p) {
    depth[u] = depth[p] + 1;
    parent[u][0] = p;
    for (int i = 1; i < LG; i++) {
        parent[u][i] = parent[parent[u][i - 1]][i - 1];
    }
    for (int v : g[u]) {
        if (v == p) continue;
        dfs(v, u);
    }
}
int lca(int u, int v) {
    if (depth[u] < depth[v]) swap(u, v);
    int diff = depth[u] - depth[v];
    for (int i = 0; i < LG; i++) {
        if (diff >> i & 1) {
            u = parent[u][i];
        }
    }
    if (u == v) return u;
    for (int i = LG - 1; i >= 0; i--) {
        if (parent[u][i] != parent[v][i]) {
            u = parent[u][i];
            v = parent[v][i];
        }
    }
    return parent[u][0];
}

```

```

    return parent[u][0];
}
int dist(int u, int v) { return depth[u] + depth[v] - 2 *
    depth[lca(u, v)]; }
int kthParent(int u, int k) {
    for (int i = 0; i < LG; i++) {
        if (k >> i & 1) {
            u = parent[u][i];
        }
    }
    return u;
}
}

7.14 SCC [57 lines] - 060054f3
vector<bool> visited; // keeps track of which vertices are
    already visited

// runs depth first search starting at vertex v.
// each visited vertex is appended to the output vector when
    dfs leaves it.
void dfs(int v, vector<vector<int>> const &adj, vector<int>
    &output) {
    visited[v] = true;
    for (auto u : adj[v])
        if (!visited[u])
            dfs(u, adj, output);
    output.push_back(v);
}

// input: adj -- adjacency list of G
// output: components -- the strongly connected components in G
// output: adj_cond -- adjacency list of G~SCC (by root
    vertices)
void scc(vector<vector<int>> const &adj, vector<vector<int>>
    &components, vector<vector<int>> &adj_cond) {
    int n = adj.size();
    components.clear(), adj_cond.clear();

    vector<int> order; // will be a sorted list of G's vertices
        by exit time

    visited.assign(n, false);

    // first series of depth first searches
    for (int i = 0; i < n; i++)
        if (!visited[i])
            dfs(i, adj, order);

    // create adjacency list of G~T
    vector<vector<int>> adj_rev(n);
    for (int v = 0; v < n; v++)
        for (int u : adj[v])
            adj_rev[u].push_back(v);

    visited.assign(n, false);
    reverse(order.begin(), order.end());

    vector<int> roots(n, 0); // gives the root vertex of a
        vertex's SCC

    // second series of depth first searches
    for (auto v : order)
        if (!visited[v]) {
            std::vector<int> component;
            dfs(v, adj_rev, component);
            components.push_back(component);
            int root = *min_element(begin(component),
                end(component));
            for (auto u : component)
                roots[u] = root;
        }

    // add edges to condensation graph
    adj_cond.assign(n, {});
    for (int v = 0; v < n; v++)

```

```

        for (auto u : adj[v])
            if (roots[v] != roots[u])
                adj_cond[roots[v]].push_back(roots[u]);
    }

7.15 SPFA O(V*E) [57 lines] - 7cd7bdcb
int q[3001] = {0}; // queue for node
int d[1001] = {0}; // record shortest path
from start to ith node bool f[1001] = {0};
int a[1001][1001] = {0}; // adjacency list
int w[1001][1001] = {0}; // adjacency matrix
int main(void) {
    int n = 0, m = 0;
    cin >> n >> m;
    for (int i = 1; i <= m; i++) {
        int x = 0, y = 0, z = 0;
        cin >> x >> y >> z;
        // node x to node y has weight z
        a[x][0]++;
        a[x][a[x][0]] = y;
        w[x][y] = z;
    }
    /*
    // for undirected graph
    a[x][0]++;
    a[y][a[y][0]] = x;
    w[y][x] = z;
    */
    int s = 0, e = 0;
    cin >> s >> e; // s: start, e: end
    SPFA(s);
    cout << d[e] << endl;
    return 0;
}

void SPFA(int v0) {
    int t, h, u, v;
    for (int i = 0; i < 1001; i++) d[i] = INT_MAX;
    for (int i = 0; i < 1001; i++) f[i] = false;
    d[v0] = 0;
    h = 0;
    t = 1;
    q[1] = v0;
    f[v0] = true;
    while (h != t) {
        h++;
        if (h > 3000) h = 1;
        u = q[h];
        for (int j = 1; j <= a[u][0]; j++) {
            v = a[u][j];
            if (d[u] + w[u][v] < d[v]) // change to > if
                calculating longest path
            {
                d[v] = d[u] + w[u][v];
                if (!f[v]) {
                    t++;
                    if (t > 3000) t = 1;
                    q[t] = v;
                    f[v] = true;
                }
            }
        }
        f[u] = false;
    }
}

7.16 Topological Sort [33 lines] - 3ddc7df6
vector<int> topologicalSort(int V, vector<vector<int>>& g) {
    vector<int> inDegree(V, 0);
    vector<int> result;
    queue<int> q;
    for (int u = 0; u < V; u++) {
        for (int v : g[u]) {
            inDegree[v]++;
        }
    }
}

```

```

    for (int i = 0; i < V; i++) {
        if (inDegree[i] == 0) {
            q.push(i);
        }
    }
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        result.push_back(u);
        for (int v : g[u]) {
            inDegree[v]--;
            if (inDegree[v] == 0) {
                q.push(v);
            }
        }
    }
    // You may need to check inDegree of all the vertices is 0
    if (result.size() != V) {
        cout << "Graph contains a cycle, topological sort not
            possible!" << endl;
        return {};
    }
    return result;
}

7.17 Tree Rerooting [99 lines] - c3cafbac
namespace reroot {
    const auto exclusive = [](const auto& a, const auto& base,
        const auto& merge_into, int vertex)
        {
            int n = (int)a.size();
            using Aggregate = std::decay_t<decltype(base)>;
            std::vector<Aggregate> b(n, base);
            for (int bit = (int)std::lg(n); bit >= 0; --bit) {
                for (int i = n - 1; i >= 0; --i) b[i] = b[i >> 1];
                int sz = n - (n & !bit);
                for (int i = 0; i < sz; ++i) {
                    int index = (i >> bit) ^ 1;
                    b[index] = merge_into(b[index], a[i], vertex, i);
                }
            }
            return b;
        };
    // MergeInto : Aggregate * Value * Vertex(int) * EdgeIndex(int)
        -> Aggregate
    // Base : Vertex(int) -> Aggregate
    // FinalizeMerge : Aggregate * Vertex(int) * EdgeIndex(int)
        -> Value
    const auto rerooter = [](const auto& g, const auto& base,
        const auto& merge_into, const auto&
        finalize_merge) {
        int n = (int)g.size();
        using Aggregate = std::decay_t<decltype(base(0))>;
        using Value = std::decay_t<decltype(finalize_merge(base(0),
            0, 0))>;
        std::vector<Value> root_dp(n), dp(n);
        std::vector<std::vector<Value>> edge_dp(n), redge_dp(n);

        std::vector<int> bfs, parent(n);
        bfs.reserve(n);
        bfs.push_back(0);
        for (int i = 0; i < n; ++i) {
            int u = bfs[i];
            for (auto v : g[u]) {
                if (parent[u] == v) continue;
                parent[v] = u;
                bfs.push_back(v);
            }
        }

        for (int i = n - 1; i >= 0; --i) {
            int u = bfs[i];

```

```

int p_edge_index = -1;
Aggregate aggregate = base(u);
for (int edge_index = 0; edge_index < (int)g[u].size();
    ++edge_index) {
    int v = g[u][edge_index];
    if (parent[u] == v) {
        p_edge_index = edge_index;
        continue;
    }
    aggregate = merge_into(aggregate, dp[v], u, edge_index);
}
dp[u] = finalize_merge(aggregate, u, p_edge_index);
}

for (auto u : bfs) {
    dp[parent[u]] = dp[u];
    edge_dp[u].reserve(g[u].size());
    for (auto v : g[u]) edge_dp[u].push_back(dp[v]);
    auto dp_exclusive = exclusive(edge_dp[u], base(u),
        merge_into, u);
    redge_dp[u].reserve(g[u].size());
    for (int i = 0; i < (int)dp_exclusive.size(); ++i) {
        auto merge_value = finalize_merge(dp_exclusive[i], u,
            i);
        redge_dp[u].push_back(merge_value);
    }
    root_dp[u] = finalize_merge(
        n > 1 ? merge_into(dp_exclusive[0], edge_dp[u][0], u,
            0) : base(u), u,
        -1);
    for (int i = 0; i < (int)g[u].size(); ++i) {
        dp[g[u][i]] = redge_dp[u][i];
    }
}

return std::make_tuple(std::move(root_dp),
    std::move(edge_dp),
    std::move(redge_dp));
};
} // namespace reroot

```

```

void solve() {
    using Aggregate = int;
    using Value = int;

    auto base = [](int vertex) -> Aggregate { return 0; };
    auto merge_into = [](Aggregate vertex_dp, Value
        neighbor_dp, int vertex,
        int edge_index) -> Aggregate {
        return vertex_dp + std::max(neighbor_dp, 0);
    };
    auto finalize_merge = [&](Aggregate vertex_dp, int vertex,
        int edge_index) -> Value {
        return vertex_dp + 2 * color[vertex] - 1;
    };

    auto [reroot_result, edge_dp, redge_dp] =
        reroot::rerooter(g, base, merge_into, finalize_merge);

    for (auto dp : reroot_result) {
        std::cout << dp << ' ';
    }
    std::cout << '\n';
}

```

8 Misc

8.1 Bit count in O(1) [5 lines] - 8f60ac14

```

int BitCount(unsigned int u) {
    unsigned int uCount;
    uCount = u - ((u >> 1) & 033333333333) - ((u >> 2) &
        011111111111);
    return ((uCount + (uCount >> 3)) & 030707070707) % 63;
}

```

8.2 Bit hacks [16 lines] - 96eab638

```

# x & -x is the least bit in x.
# iterate over all the subsets of the mask
for (int s=m; ; s=(s-1)&m) {
    ... you can use s ...
    if (s==0) break;
}
# c = x&-x, r = x+c; (((r^x) >> 2)/c) | r is the
next number after x with the same number of bits set.
# __builtin_popcount(x) //number of ones in binary
__builtin_popcountll(x) // for long long
# __builtin_clz(x) // number of leading zeros
__builtin_ctz(x) // number of trailing zeros, they also have
long long version
# bitset<N> b; //N is size of bitset
b.count(); // return number of bits set 0(N / 64)
b.to_string(); // return string representation of bitset
b.any() // return true if any bit is set 0(N / 64)

```

8.3 Bitset C++ [13 lines] - 28766a3c

```

bitset<17>BS;
BS[1] = BS[7] = 1;
cout<<BS._Find_first()<<endl; // prints 1
bs._Find_next(idx). This function returns first set bit after
index idx.for example:

bitset<17>BS;
BS[1] = BS[7] = 1;
cout<<BS._Find_next(1)<<','<<BS._Find_next(3)<<endl; // prints
7,7
So this code will print all of the set bits of BS:

for(int i=BS._Find_first();i< BS.size();i = BS._Find_next(i))
    cout<<i<<endl;
//Note that there isn't any set bit after idx,
BS._Find_next(idx) will return BS.size(); same as calling
BS._Find_first() when bitset is clear;

```

8.4 Knight Moves [2 lines] - 6c4e6fbc

```

int X[8]={2,1,-1,-2,-2,-1,1,2};
int Y[8]={1,2,2,1,-1,-2,-2,-1};

```

8.5 Random Generator [6 lines] - 2b02ac44

```

#define accuracy
chrono::steady_clock::now().time_since_epoch().count()
mt19937 rng(accuracy);
int rand(int l, int r) {
    uniform_int_distribution<int> ludo(l, r);
    return ludo(rng);
}

```

8.6 Special DS [14 lines] - 0cb955b1

```

#include<ext/pb_ds/assoc_container.hpp>
#include<ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
template <typename T> using o_set = tree<T, null_type,
    less<T>, rb_tree_tag, tree_order_statistics_node_update>;
// o_set<int> st;
// *(st.find_by_order(index))
// st.order_of_key(value)

// Faster hash
gp_hash_table<int, int> mp;
// Or
unordered_map<int, int> mp;
mp.reserve(n * 2);
mp.max_load_factor(0.7);

```

8.7 Usefull Things [9 lines] - 2bbfead1

```

C++:
#pragma GCC optimize("Ofast")
#pragma GCC target("avx,avx2,fma")
ios_base::sync_with_stdio(false), cin.tie(nullptr),
    cout.tie(nullptr);
freopen("input.txt", "r", stdin);
->Python:

```

```

import sys
input = sys.stdin.readline
sys.stdout.write("-----")

```

8.8 python syntax [331 lines] - 623b4852

```

1. Variables
# int a=5; long long b=...;
a = 5
b = 10**18 # arbitrary-size int
x = 3.5 # float
flag = True # True / False
s = "abc"
c = 'a' # str of length 1
MOD = 10**9 + 7 # integer
INF = float('inf')
a, b = b, a # swap
a += 1 # no a++ or a--
# 1e9+7 is float; use 10**9+7.
# Integers use memory; floats are not exact.
# Use straight quotes: 'abc' or "abc".
-----

```

```

2. Input / Output
import sys
input = sys.stdin.readline
n = int(input())
a, b = map(int, input().split())
v = list(map(int, input().split()))
s = input().rstrip('\n') # newline only
s = input().strip() # edge whitespace
g = [list(map(int, input().split()))
    for _ in range(n)]
# Alternative: read ALL remaining input.
data = sys.stdin.read().split()
print(a)
print(a, b)
print(*v) # space-separated
print(*v, sep="\n")
print("YES" if ok else "NO")
print(f"{x:.6f}") # fixed, 6 decimals
print("\n".join(map(str, out)))
# C++ equivalent: fixed << setprecision(6)
# readline() includes newline, EOF gives ''.
-----

```

```

3. Containers
v = [] # vector
v = [0] * n
v = [1, 2, 3]
g = [[0]*m for _ in range(n)]
# NOT [[0]*m]*n: rows would be shared.
adj = [[] for _ in range(n)]
p = (1, 2) # tuple / pair
x, y = p
st = set() # unordered_set
st = {1, 2, 3}
mp = {} # dict, NOT set
mp = {1: 10, 2: 20}
from collections import Counter
from collections import defaultdict, deque
cnt = Counter()
d = defaultdict(int) # default 0
adj = defaultdict(list) # default []
q = deque([1, 2, 3])
stack = [] # append / pop
import heapq
h = [] # min-heap
-----

```

4. Control flow / Functions

```

if a > b:
    pass
elif a == b:
    pass
else:
    pass

```

```
# && / || / ! -> and / or / not
for i in range(n): # 0 .. n-1
    pass
for i in range(1, n+1): # 1 .. n
    pass
for i in range(n-1, -1, -1):
    pass # n-1 .. 0
for i in range(0, n, 2):
    pass
for x in v:
    pass
for i, x in enumerate(v):
    pass
for x, y in zip(a, b):
    pass # shorter length
while cond:
    if stop:
        break
    if skip:
        continue
    # Other work must update loop state.
def f(a, b):
    return a + b
def g():
    return 1, 2
x, y = g()
add = lambda a, b: a + b
ans = x if cond else y
def h():
    global cnt # reassign global
    cnt += 1
def main():
    pass
main() # local vars often faster
-----
5. List operations
v[0] # first
v[-1] # last
v[1:3] # indices 1, 2
v[::-1] # reversed copy
len(v)
v.append(x) # amortized O(1)
v.pop() # last; O(1)
v.pop(i) # worst O(n)
v.insert(i, x) # O(n)
v.remove(x) # first match; O(n)
v.extend(w) # append all of w
v + w # new concatenated list
v.index(x) # first index; O(n)
v.count(x) # frequency; O(n)
x in v # membership; O(n)
v.sort() # in-place; returns None
v.sort(reverse=True)
v.reverse() # in-place
w = v[:] # shallow copy
# Nested mutable items remain shared.
del v[i]
v.clear()
sum(v)
min(v)
max(v)
sq = [i*i for i in range(5)]
ev = [x for x in v if x % 2 == 0]
# index/remove: ValueError if not found.
# pop/access: IndexError if invalid.
# min/max on empty list: ValueError.
-----
6. Basic functions / Division
abs(-5) # 5
min(3, 7) # 3
max(3, 7) # 7
sum([1, 2, 3]) # 6
len("abc") # 3
pow(2, 10) # 1024
```

```
pow(2, 10, 1000) # 24; modular power
pow(a, -1, p) # modular inverse
# Python 3.8+; p > 1 and gcd(a,p) == 1.
# Otherwise inverse raises ValueError.
divmod(17, 5) # (3, 2)
17 // 5 # 3; floor division
17 % 5 # 2
-7 // 2 # -4; C++ gives -3
-7 % 3 # 2; C++ gives -1
17 / 5 # 3.4; float
# // floors; C++ int division truncates to 0.
# Python remainder follows divisor's sign.
all(v) # every item truthy
any(v) # any item truthy
# all([]) == True; any([]) == False
list(range(5)) # [0,1,2,3,4]
list(map(int, ["1", "2"])) # [1,2]
sorted(v) # new sorted list
list(reversed(v)) # reversed copy
-----
7. Mathematics
import math
from math import gcd, lcm, isqrt
from math import comb, factorial
gcd(12, 18) # 6
gcd(12, 18, 24) # 6; Python 3.9+
lcm(4, 6) # 12; Python 3.9+
# Older versions, for positive a,b:
a // gcd(a, b) * b
math.sqrt(16) # 4.0; float
isqrt(17) # 4; exact int sqrt
math.floor(2.7) # 2
math.ceil(2.1) # 3
factorial(5) # 120
comb(5, 2) # 10; nCr
math.log(8, 2) # 3.0
math.log2(8)
math.log10(100)
math.inf
math.pi
(a + b - 1) // b # ceil(a/b), b > 0
-(a // b) # ceil(a/b), b != 0
r = isqrt(n) # n >= 0
r*r == n # perfect square?
# isqrt and comb: Python 3.8+.
# Avoid float sqrt/log for exact large ints.
-----
8. Strings
s[0]
s[-1]
s[1:3]
s[::-1]
len(s)
s.split() # whitespace split
"a,b".split(",") # ['a','b']
" ".join(["a", "b"]) # 'a b'
"".join(chars)
s.strip() # edge whitespace
s.replace("l", "L") # returns new string
s.find("l") # index or -1
s.count("l") # non-overlapping
s.startswith("he")
s.endswith("lo")
s.lower()
s.upper()
s.isdigit()
s.isalpha()
s.islower()
s.isupper()
ord('c') - ord('a') # 2
chr(ord('a') + 2) # 'c'
# Strings are immutable: s[0]='x' fails.
chars = list(s)
chars[0] = 'x' # requires nonempty s
```

```
s = "".join(chars)
# isdigit/isalpha are Unicode-aware.
# String methods above do not modify s.
-----
9. Set / Dictionary
st.add(x)
st.remove(x) # KeyError if absent
st.discard(x) # safe if absent
x in st # average O(1)
len(st) # O(1)
a | b # union (sets)
a & b # intersection
a - b # difference
a ^ b # symmetric difference
# Sets are unordered; no lower_bound.
sorted(st) # sorted list; O(n log n)
d[k] = value
d[k] # KeyError if absent
d.get(k, 0) # default, no insertion
k in d # checks KEYS
del d[k] # KeyError if absent
d.pop(k, None) # safe removal
for k, value in d.items():
    pass
d.keys() # view of keys
d.values() # view of values
{i: i*i for i in range(3)}
# dict: insertion order, NOT sorted key order.
# Set/dict keys must be hashable (not lists).
# Hash lookup: avg O(1), worst O(n).
-----
10. Counter / defaultdict / deque
from collections import Counter
from collections import defaultdict, deque
c = Counter(v)
c[x] += 1
c[x] -= 1
c[missing] # 0; no key inserted
c.most_common(1) # [(value, count)]
del c[x]
# Zero count does not automatically delete key.
# Counter can store zero/negative counts.
g = defaultdict(list)
g[u].append(v) # missing g[u] -> []
d = defaultdict(int)
d[x] += 1 # missing d[x] -> 0
# d.get(x) does NOT invoke default factory.
q = deque()
q.append(x) # push back; O(1)
q.appendleft(x) # push front; O(1)
q.pop() # pop back; O(1)
q.popleft() # pop front; O(1)
q[0] # front; O(1)
q[-1] # back; O(1)
len(q)
if q:
    pass # nonempty check
# Empty pop/front/back raises IndexError.
# deque middle indexing: O(n).
-----
11. Sort / Heap / Binary search
v.sort() # ascending; O(n log n)
v.sort(reverse=True)
v.sort(key=lambda p: p[1]) # second field
v.sort(key=lambda p: (-p[0], p[1]))
# Python sorting is stable.
import heapq
h = []
heapq.heappush(h, x) # O(log n)
heapq.heappop(h) # smallest; O(log n)
h[0] # smallest; O(1)
heapq.heapify(v) # in-place; O(n)
# v is a heap afterwards, NOT a sorted list.
h = [] # numeric max-heap
```

```

heapq.heappush(h, -x)
-heapq.heappop(h) # real maximum
-h[0] # real maximum at top
h = [] # tuple min-heap
heapq.heappush(h, (dist, node))
# Tuples compare lexicographically.
from bisect import bisect_left, bisect_right
bisect_left(v, x) # first index with >= x
bisect_right(v, x) # first index with > x
# Binary search requires ASCENDING sorted v.
# Returns index in [0,len(v)]; 0(log n).
# bisect is NOT valid on a heap array.
# Empty heappop / h[0]: IndexError.
-----

```

```

12. Bit operations
x & y # AND
x | y # OR
x ^ y # XOR
~x # NOT: -x-1
x << k # shift left
x >> k # arithmetic shift right
(x >> i) & 1 # check bit i
x | (1 << i) # value with bit set
x ^ (1 << i) # value with bit flipped
x & ~(1 << i) # value with bit cleared
# Assign result to x to actually update x.
x |= 1 << i
x ^= 1 << i
x &= ~(1 << i)
x & -x # lowest set bit; 0 if x=0
x.bit_count() # Python 3.10+
bin(x).count("1") # older Python
# Both count bits in abs(x) for negative x.
# Python ints have no fixed bit width.
# Emulate a w-bit unsigned complement:
(~x) & ((1 << w) - 1)
# Shift counts / bit indices must be >= 0.
for mask in range(1 << n): # n >= 0; all subsets
    pass
# Bit operations on large ints cost more.
# Use nonnegative masks for usual CP tasks

```

9 Number Theory

9.1 10-ary to m-ary [22 lines] - e6c26ef1

```

char a[16]={ '0','1','2','3','4','5','6','7',
             '8','9','A','B','C','D','E','F' };
string tenToM(int n, int m){
    int temp=n;
    string result="";
    while (temp!=0){
        result=a[temp%10]+result;
        temp/=10;
    }
    return result;
}
// m-ary to 10-ary
string num = "0123456789ABCDE";
int mToTen(string n, int m){
    int multi=1;
    int result=0;
    for (int i=n.size()-1; i>=0; i--) {
        result += num.find(n[i])*multi;
        multi*=m;
    }
    return result;
}

```

9.2 Binary Exponentiation [9 lines] - 8b74ed31

```

int power(long long n, long long k, int mod) {
    int ans = 1 % mod; n %= mod; if (n < 0) n += mod;
    while (k) {
        if (k & 1) ans = (long long) ans * n % mod;
        n = (long long) n * n % mod;
        k >>= 1;
    }
}

```

```

}
return ans;
}

```

9.3 CRT [21 lines] - 6dfeca05

```

ll egcd(ll a, ll b, ll &x, ll &y){
    if(b == 0) { x=1, y=0; return a;}
    ll g = egcd(b, a%b, y, x);
    y -= a/b*x; return g;
}
ll crt(ll r1, ll m1, ll r2, ll m2){
    if(m1<m2) swap(r1, r2), swap(m1, m2);
    ll p, q, g = egcd(m1, m2, p, q);
    if((r2-r1)%g !=0) return -1; //no solution
    ll x = (r2-r1)%m2*p/m2*m1/g + r1;
    return x<0? x+m1*m2/g: x;
}
ll crt(vector<ll>& r, vector<ll>& m){
    ll x = r[0], M=m[0];
    for (int i = 1; i < r.size(); ++i){
        x = crt(x, M, r[i], m[i]);
        ll g = __gcd(M, m[i]);
        M = (M/g)*m[i];
    }
    return x;
}

```

9.4 Catalan Numbers [8 lines] - 17405386

```

void catalan(int n) {
    int res = 1;
    cout << res << " ";
    for (int i = 1; i < n; i++) {
        res = (res * (4 * i - 2)) / (i + 1);
        cout << res << " ";
    }
}

```

9.5 Count 1s 0 to n [9 lines] - 6b5041db

```

int cntOnes(int n) {
    int cnt = 0;
    for (int i = 1; i <= n; i <= 1) {
        int x = (n + 1) / (i << 1);
        cnt += x * i;
        if ((n + 1) % i && n & i) cnt += (n + 1) % i;
    }
    return cnt;
}

```

9.6 Divisors multiple info [36 lines] - 3aff6f8b

```

constexpr int N = 1000005;

```

```

void Divisors(int n) {
    int sum = 1, total = 1;
    int mnP = INT_MAX, mxP = INT_MIN, cntP = 0, totalP = 0;
    for (int i = 0; i < N && Prime[i] * Prime[i] <= n; i++) {
        if (n % Prime[i] == 0) {
            mnP = min(mnP, Prime[i]);
            mxP = max(mxP, Prime[i]);
            int k = 0;
            cntP++;
            while (n % Prime[i] == 0) {
                k++;
                n /= Prime[i];
            }
            sum *= (k + 1); // NOD
            totalP += k;
            int s = 0, p = 1;
            while (k-- >= 0) {
                s += p;
                p *= Prime[i];
            };
            total *= s; // SOD
        }
    }
    if (n > 1) {

```

```

        cntP++, totalP++;
        sum *= 2;
        total *= (1 + n);
        mnP = min(mnP, n);
        mxP = max(mxP, n);
    }
    cout << mnP << " " << mxP << " " << cntP << " " << totalP <<
        " " << sum << " "
        << total << "\n";
}

```

9.7 Extended GCD [8 lines] - 5998ac10

```

// return {x,y} such that ax+by=gcd(a,b)
int exgcd(int a, int b, int &x, int &y) {
    if (b == 0) {
        x = 1; y = 0; return a; }
    int g = exgcd(b, a % b, y, x);
    y -= a / b * x;
    return g;
} // bezout's identity: a.x + b.y = gcd(a,b)

```

9.8 Find numbers in L to R divisible by all array elements [12 lines] - 12567a3c

```

solve(int* arr, int N, int L, int R) {
    int LCM = arr[0];
    for (int i = 1; i < N; i++) {
        LCM = (LCM * arr[i]) / (__gcd(LCM, arr[i]));
    }
    if ((LCM < L && LCM * 2 > R) || LCM > R) {
        return;
    }
    int k = (L / LCM) * LCM;
    if (k < L) k = k + LCM;
    for (int i = k; i <= R; i = i + LCM) cout << i << ' ';
}

```

9.9 GaussElimination [39 lines] - 8a474ce2

```

template<typename ld>
int gauss(vector<vector<ld>>& a, vector<ld>& ans) {
    const ld EPS = 1e-9;
    int n = a.size(); //number of equations
    int m = a[0].size() - 1; //number of variables
    vector<int> where(m, -1); //indicates which row contains the
        solution
    int row, col;
    for (col = 0, row = 0; col < m && row < n; ++col) {
        int sel = row; //which row contains the maximum value/
        for (int i = row + 1; i < n; i++)
            if (abs(a[i][col]) > abs(a[sel][col]))
                sel = i;
        if (abs(a[sel][col]) < EPS) continue; //it's basically 0.
        a[sel].swap(a[row]); //taking the max row up
        where[col] = row;
        ld t = a[row][col];
        for (int i = col; i <= m; i++) a[row][i] /= t;
        for (int i = 0; i < n; i++) {
            if (i != row) {
                ld c = a[i][col];
                for (int j = col; j <= m; j++)
                    a[i][j] -= a[row][j] * c;
            }
        }
        row++;
    }
    ans.assign(m, 0);
    for (int i = 0; i < m; i++)
        if (where[i] != -1)
            ans[i] = a[where[i]][m] / a[where[i]][i];
    for (int i = 0; i < n; i++) {
        ld sum = 0;
        for (int j = 0; j < m; j++)
            sum += ans[j] * a[i][j];
        if (abs(sum - a[i][m]) > EPS) //L.H.S!=R.H.S
            ans.clear(); //No solution
    }
}

```

```
    return row;
}
```

9.10 GaussMod2 [43 lines] - af4935b7

```
template<typename T>
struct Gauss {
    int bits;
    vector<T> table;
    int rank = 0; // track number of basis/independent
                  vectors
    Gauss() : bits(60) { table = vector<T>(bits, 0); }
    Gauss(int _bits) : bits(_bits) { table = vector<T>(bits,
        0); }
    // Return rank/size of basis
    int basis_size() { return rank; }
    // Check if x can be obtained from the basis
    bool can(T x) {
        for (int i = bits - 1; i >= 0; i--) {
            if (!(x >> i & 1)) continue;
            if (!table[i]) return false;
            x ^= table[i];
        }
        return true;
    }
    void add(T x) {
        for (int i = bits - 1; i >= 0; i--) {
            if (!(x >> i & 1)) continue; // Skip if bit i is
            0
            if (!table[i]) {
                table[i] = x;
                rank++;
                return;
            }
            x ^= table[i]; // Eliminate this bit using XOR
        }
        // x became 0 (linearly dependent)
    }
    T get_max_xor() {
        T ans = 0;
        for (int i = bits - 1; i >= 0; i--) {
            if ((ans ^ table[i]) > ans) {
                ans ^= table[i];
            }
        }
        return ans;
    }
};
// Gauss<long long> g(60);
// g.add(x);
// g.get_max_xor();
```

9.11 Largest Power of n! [11 lines] - ffed3d40

```
// Returns largest power of p that divides n!
int largestPower(int n, int p) {
    int res = 0;

    // Calculate res = n/p + n/(p^2) + n/(p^3) + ....
    while (n > 0) {
        n /= p;
        res += n;
    }
    return res;
}
```

9.12 Leap Year [13 lines] - 8ff2b7cb

```
bool isLeap(int n){
    if (n%100==0)
        if (n%400==0) return true;
        else return false;
    if (n%4==0) return true;
    else return false;
}
// Num of Leap year in between
int calNum(int year) {
    return (year / 4) - (year / 100) +
```

```
(year / 400);
}
int leapNum(int l, int r) { return calNum(r) - calNum(--l);}
```

9.13 Logab [3 lines] - 1b0f688c

```
int logab (int a, int b){
    return log2(a) / log2(b);
}
```

9.14 Matrix [105 lines] - 0cf635bb

```
template<typename T>
struct Matrix {
    T MOD = 1e9 + 7; //change if necessary
    T add(T a, T b) const {
        T res = a + b;
        if (res >= MOD) return res - MOD;
        return res;
    }
    T sub(T a, T b) const {
        T res = a - b;
        if (res < 0) return res + MOD;
        return res;
    }
    T mul(T a, T b) const {
        T res = a * b;
        if (res >= MOD) return res % MOD;
        return res;
    }
    int R, C;
    vector<vector<T>>mat;
    Matrix(int _R = 0, int _C = 0) {
        R = _R, C = _C;
        mat.resize(R);
        for (auto& v : mat) v.assign(C, 0);
    }
    void print() {
        for (int i = 0; i < R; i++)
            for (int j = 0; j < C; j++)
                cout << mat[i][j] << " \n"[j == C - 1];
    }
    void createIdentity() {
        for (int i = 0; i < R; i++)
            for (int j = 0; j < C; j++)
                mat[i][j] = (i == j);
    }
    Matrix operator+(const Matrix& o) const {
        Matrix res(R, C);
        for (int i = 0; i < R; i++)
            for (int j = 0; j < C; j++)
                res[i][j] = add(mat[i][j] + o.mat[i][j]);
    }
    Matrix operator-(const Matrix& o) const {
        Matrix res(R, C);
        for (int i = 0; i < R; i++)
            for (int j = 0; j < C; j++)
                res[i][j] = sub(mat[i][j] + o.mat[i][j]);
    }
    Matrix operator*(const Matrix& o) const {
        Matrix res(R, o.C);
        for (int i = 0; i < R; i++)
            for (int j = 0; j < o.C; j++)
                for (int k = 0; k < C; k++)
                    res.mat[i][j] = add(res.mat[i][j], mul(mat[i][k],
                        o.mat[k][j]));
        return res;
    }
    Matrix pow(long long x) {
        Matrix res(R, C);
        res.createIdentity();
        Matrix<T> o = *this;
        while (x) {
            if (x & 1) res = res * o;
            o = o * o;
            x >>= 1;
        }
    }
}
```

```
    return res;
}
Matrix inverse()//Only square matrix && non-zero
determinant
{
    Matrix res(R, R + R);
    for (int i = 0; i < R; i++) {
        for (int j = 0; j < R; j++)
            res.mat[i][j] = mat[i][j];
        res.mat[i][R + i] = 1;
    }
    for (int i = 0; i < R; i++) {
        //find row 'r' with highest value at [r][i]
        int tr = i;
        for (int j = i + 1; j < R; j++)
            if (abs(res.mat[j][i]) > abs(res.mat[tr][i]))
                tr = j;
        //swap the row
        res.mat[tr].swap(res.mat[i]);
        //make 1 at [i][i]
        T val = res.mat[i][i];
        for (int j = 0; j < R + R; j++) res.mat[i][j] /= val;
        //eliminate [r][i] from every row except i.
        for (int j = 0; j < R; j++) {
            if (j == i) continue;
            for (int k = R + R - 1; k >= i; k--) {
                res.mat[j][k] -= res.mat[i][k] * res.mat[j][i] /
                    res.mat[i][i];
            }
        }
    }
    Matrix ans(R, R);
    for (int i = 0; i < R; i++)
        for (int j = 0; j < R; j++)
            ans.mat[i][j] = res.mat[i][R + j];
    return ans;
}
// Matrix<long long> base(2, 2);
// base.mat[0][0] = 1LL; base.mat[0][1] = 1LL;
// base.mat[1][0] = 1LL; base.mat[1][1] = 0LL;
// Matrix<long long> result = base.pow(n - 1);
// cout << result.mat[0][0] << '\n'; // nth Fibonacci number

9.15 Miller-Rabin-Pollard-Rho [67 lines] - 05b017db
using u64 = uint64_t;
using u128 = __uint128_t;

u64 binpower(u64 b, u64 e, u64 mod) {
    u64 r = 1;
    b %= mod;
    while (e) {
        if (e & 1) r = (u128)r * b % mod;
        b = (u128)b * b % mod;
        e >>= 1;
    }
    return r;
}

bool check_composite(u64 n, u64 a, u64 d, int s) {
    u64 x = binpower(a, d, n);
    if (x == 1 || x == n - 1) return false;
    for (int r = 1; r < s; r++) {
        x = (u128)x * x % n;
        if (x == n - 1) return false;
    }
    return true;
}

// These bases make the test deterministic for every n < 2^64.
bool MillerRabin(u64 n) {
    if (n < 2) return false;
    int s = 0;
    u64 d = n - 1;
    while ((d & 1) == 0) {
```

```

    d >= 1;
    s++;
}
for (u64 a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}) {
    if (n == a) return true;
    if (check_composite(n, a, d, s)) return false;
}
return true;
}

// One non-trivial factor of a composite n. Floyd cycle finding
// on
// x -> x^2 + c; a c that only finds n itself is retried with
// the next one.
u64 pollard(u64 n) {
    if (n % 2 == 0) return 2;
    for (u64 c = 1;; c++) {
        auto f = [&](u64 x) { return ((u128)x * x + c) % n; };
        u64 x = 2, y = 2, d = 1;
        while (d == 1) {
            x = f(x);
            y = f(f(y));
            d = __gcd(x > y ? x - y : y - x, n);
        }
        if (d != n) return d;
    }
}

// Full factorisation, unsorted. sort(out.begin(), out.end())
// if it matters.
void factor(u64 n, vector<u64>& out) {
    if (n == 1) return;
    if (MillerRabin(n)) {
        out.push_back(n);
        return;
    }
    u64 d = pollard(n);
    factor(d, out);
    factor(n / d, out);
}

```

9.16 Mobius [12 lines] - a32332ff

```

const int N = 1e6 + 5;
int mob[N];
void mobius() {
    mob[1] = 1;
    for (int i = 2; i < N; i++){
        mob[i]--;
        for (int j = i + i; j < N; j += i) {
            mob[j] -= mob[i];
        }
    }
}

```

9.17 Mod Inverse [5 lines] - 6e1326cf

```

int modInv(int a, int m) {
    int x, y; //if g==1 Inverse doesn't exist
    int g = gcdExt(a, m, x, y);
    return (x % m + m) % m;
}

```

9.18 Modular Operations [12 lines] - da7829be

```

int add(int x, int y) {
    x += y;
    if (x >= mod) x -= mod;
    if (x < 0) x += mod;
    return x;
}
int mul(int x, int y) {
    return x * 1LL * y % mod;
}
int divide(int x, int y) {
    return mul(x, power(y, mod - 2));
}

```

9.19 NOD of n! [13 lines] - e777f444

```

ull factorialDivisors(ull n) {
    ull result = 1;
    for (int i = 0; i < allPrimes.size(); i++) {
        ull p = allPrimes[i];
        ull exp = 0;
        while (p <= n) {
            exp = exp + (n / p);
            p = p * allPrimes[i];
        }
        result = result * (exp + 1);
    }
    return result;
}

```

9.20 No of Digits in n! in base B [7 lines] - 1f8efcc5

```

ll NoOfDigitInNFactInBaseB(ll N,ll B){
    ll i;
    double ans=0;
    for(i=1;i<=N;i++)ans+=log(i);
    ans=ans/log(B),ans=ans+1;
    return (ll)ans;
}

```

9.21 Principle of Inclusion Exclusion [21 lines] - cb2ccfa4

```

// count the numbers between 1 and n (inclusive) that are not
// divisible by any
// of the integers in the given array a
// is n - (div by one number - div by two number + div by three
// number - ...)
int PIE(int div[], int n, int num) {
    int sum = 0;
    for (int msk = 1; msk < (1 << n); ++msk) {
        int bit_cnt = 0;
        int cur_lcm = 1;
        for (int i = 0; i < n; ++i) {
            if (msk & (1 << i)) {
                ++bit_cnt;
                cur_lcm = LCM(cur_lcm, div[i]);
            }
        }
        int cur = num / cur_lcm;
        if (bit_cnt & 1) sum += cur;
        else sum -= cur;
    }
    return num - sum;
}

```

9.22 Print Calander of Any Year [44 lines] - 1a53cdae

```

int dayNumber(int day, int month, int year) {
    static int t[] = {0, 3, 2, 5, 0, 3, 5, 1, 4, 6, 2, 4};
    year -= month < 3;
    return (year + year / 4 - year / 100 + year / 400 + t[month
        - 1] + day) % 7;
}
string getMonthName(int monthNumber) {
    string months[] = {"January", "February", "March",
        "April", "May", "June", "July",
        "August", "September", "October", "November",
        "December"};
    return (months[monthNumber]);
}
int numberOfDays(int monthNumber, int year) {
    if (monthNumber == 1 && isLeapYear(year)) return 29;
    int monthDays[] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31,
        30, 31};
    return (monthDays[monthNumber]);
}
void printCalendar(int year) {
    printf("        Calendar - %d\n\n", year);
    int days;
    int current = dayNumber(1, 1, year);
}

```

```

// i/-> Iterate through all the months
// j/-> Iterate through all the days of
the month - i for (int i = 0; i < 12; i++) {
    days = numberOfDays(i, year);
    cout << "        |" << getMonthName(i).c_str() << "|" <<
        endl;
    printf(" Sun Mon Tue Wed Thu Fri
        Sat\n");
    int k;
    for (k = 0; k < current; k++)
        printf(" ");
    for (int j = 1; j <= days; j++) {
        printf("%4d", j);
        if (++k > 6) {
            k = 0;
            cout << endl;
        }
        if (k)
            cout << endl;
        cout <<
            "-----\n";
        current = k;
    }
}

```

9.23 SOD, NOD Upto N [30 lines] - aa4e95d9

```

int SOD_upto_N(int N) { // sqrt(N)
    int res = 0;
    for (int i = 1; i <= N; i++) {
        int j = N / (N / i); // Last number with same quotient
        res += (N / i) * (i + j) * (j - i + 1) / 2;
        i = j;
    }
    return res;
}
int NOD_upto_N(ll N) { // sqrt(N)
    int res = 0;
    int u = sqrt(N);
    for (int i = 1; i <= u; i++) {
        res += (N / i) - i;
    }
    res = res * 2 + u;
    return res;
}
void NOD_per_number_upto_N(int n) { // n loglog(n)
    for (int i = 1; i <= n; i++) {
        for (int j = i; j <= n; j += i) {
            d[j]++;
            // d[j] += i // for sum of divisors
        }
    }
    for (int i = 1; i <= n; i++) {
        cout << d[i] << ' ';
    }
    return 0;
}

```

9.24 SPF [10 lines] - 802defe6

```

int spf[N];
void buildSpf() {
    for (int i = 2; i < N; i++) {
        spf[i] = i;
    }
    for (int i = 2; 1LL * i * i < N; i++)
        if (spf[i] == i)
            for (int j = i * i; j < N; j += i)
                if (spf[j] == j) spf[j] = i;
}

```

9.25 Segmented Sieve [17 lines] - b1ac5ad7

```
void segmentedSieve(int L, int R) {
    bool isPrime[R - L + 1];
    for (int i = 0; i <= R - L + 1; i++) isPrime[i] = true;
    if (L == 1) isPrime[0] = false;
    for (int i = 0; primes[i] * primes[i] <= R; i++) {
        int curPrime = primes[i];
        int base = curPrime * curPrime;
        if (base < L) {
            base = ((L + curPrime - 1) / curPrime) * curPrime;
        }
        for (int j = base; j <= R; j += curPrime) isPrime[j - L]
            = false;
    }
    for (int i = 0; i <= R - L; i++) {
        if (isPrime[i] == true) cout << L + i << '\n';
    }
    cout << '\n';
}
```

9.26 Sieve Phi Mobius [13 lines] - fd92a8a5

```
const int N = 5000005;
int phi[N];
unsigned long long phiSum[N];
void phiCalc() {
    for (int i = 1; i < N; i++) phi[i] = i;
    for (int i = 2; i < N; i++) {
        if (phi[i] == i) {
            for (int j = i; j < N; j += i) {
                phi[j] -= phi[j] / i;
            }
        }
        for (int i = 2; i < N; i++) {
            phiSum[i] = (unsigned long long)phi[i] * (unsigned long
                long)phi[i] + phiSum[i - 1];
        }
    }
}
```

9.27 Sieve upto 1e9 [71 lines] - 5fa0292a

```
vector<int> sieve(const int N, const int Q = 17, const int L
    = 1 << 15) {
    static const int rs[] = {1, 7, 11, 13, 17, 19, 23, 29};
    struct P {
        P(int p) : p(p) {}
        int p; int pos[8];
    };
    auto approx_prime_count = [] (const int N) -> int {
        return N > 60184 ? N / (log(N) - 1.1)
            : max(1., N / (log(N) - 1.11)) + 1;
    };
    const int v = sqrt(N), vv = sqrt(v);
    vector<bool> isp(v + 1, true);
    for (int i = 2; i <= vv; ++i) if (isp[i]) {
        for (int j = i * i; j <= v; j += i) isp[j] = false;
    }
    const int rsize = approx_prime_count(N + 30);
    vector<int> primes = {2, 3, 5}; int psize = 3;
    primes.resize(rsize);

    vector<P> sprimes; size_t pbeg = 0;
    int prod = 1;
    for (int p = 7; p <= v; ++p) {
        if (!isp[p]) continue;
        if (p <= Q) prod *= p, ++pbeg, primes[psize++] = p;
        auto pp = P(p);
        for (int t = 0; t < 8; ++t) {
            int j = (p <= Q) ? p : p * p;
            while (j % 30 != rs[t]) j += p << 1;
            pp.pos[t] = j / 30;
        }
        sprimes.push_back(pp);
    }

    vector<unsigned char> pre(prod, 0xFF);
    for (size_t pi = 0; pi < pbeg; ++pi) {
        auto pp = sprimes[pi]; const int p = pp.p;
        for (int t = 0; t < 8; ++t) {
```

```
        const unsigned char m = ~(1 << t);
        for (int i = pp.pos[t]; i < prod; i += p) pre[i] &= m;
    }
}

const int block_size = (L + prod - 1) / prod * prod;
vector<unsigned char> block(block_size); unsigned char*
    pblock = block.data();
const int M = (N + 29) / 30;

for (int beg = 0; beg < M; beg += block_size, pblock -=
    block_size) {
    int end = min(M, beg + block_size);
    for (int i = beg; i < end; i += prod) {
        copy(pre.begin(), pre.end(), pblock + i);
    }
    if (beg == 0) pblock[0] &= 0xFE;
    for (size_t pi = pbeg; pi < sprimes.size(); ++pi) {
        auto& pp = sprimes[pi];
        const int p = pp.p;
        for (int t = 0; t < 8; ++t) {
            int i = pp.pos[t]; const unsigned char m = ~(1 << t);
            for (; i < end; i += p) pblock[i] &= m;
            pp.pos[t] = i;
        }
    }
    for (int i = beg; i < end; ++i) {
        for (int m = pblock[i]; m > 0; m &= m - 1) {
            primes[psize++] = i * 30 + rs[__builtin_ctz(m)];
        }
    }
    assert(psize <= rsize);
    while (psize > 0 && primes[psize - 1] > N) --psize;
    primes.resize(psize);
    return primes;
}
```

9.28 Sieve [17 lines] - bc172e7b

```
const int N = 1e7 + 3;
vector<int> primes;
int notprime[N];
void sieve() {
    primes.push_back(2);
    for (int i = 4; i < N; i += 2) {
        notprime[i] = true;
    }
    for (int i = 3; i < N; i += 2) {
        if (!notprime[i]) {
            primes.push_back(i);
            for (int j = i * i; j < N; j += 2 * i) {
                notprime[j] = true;
            }
        }
    }
}
```

9.29 String Mod [5 lines] - fce25f9e

```
int mod(string& num, int a) {
    int res = 0;
    for (int i = 0; i < num.length(); i++) res = (res * 10 +
        num[i] - '0') % a;
    return res;
}
```

9.30 euler's power [17 lines] - 182b5213

```
a^n=a^(n mod phi(m)) (mod m) if gcd(a,m)=1

int phi(int n) {
    int ans = n;
    for (int i = 2; 1LL * i * i <= n; i++) {
        if (n % i == 0) {
            while (n % i == 0) n /= i;
            ans -= ans / i;
        }
    }
}
```

```
    }
    if (n > 1) ans -= ans / n;
    return ans;
}

int euler_power(int a, long long n, int m) {
    return binpow(a, n % phi(m), m);
}

9.31 nCr [39 lines] - 9bd950f6
const int N = 1e6 + 5, mod = 1e9 + 7;
int fact[N], ifact[N];

void preFact() {
    fact[0] = 1;
    for (int i = 1; i < N; i++) {
        fact[i] = 1LL * fact[i - 1] * i % mod;
    }
    ifact[N - 1] = power(fact[N - 1], mod - 2);
    for (int i = N - 2; i >= 0; i--) {
        ifact[i] = 1LL * ifact[i + 1] * (i + 1) % mod;
    }
}

int nCr(int n, int r) {
    if (n < r || n < 0) return 0;
    return 1LL * fact[n] * ifact[r] % mod * ifact[n - r] %
        mod;
}

// more space less time
const int MOD = 1e9 + 7;
const int MAX = 1e7+10;
vector<int> fact(MAX), inv(MAX);
void factorial(){
    fact[0] = 1;
    for (int i = 1; i < MAX; i++)
        fact[i] = (i * fact[i - 1]) % MOD;
}

binaryExp(int a, int n, int M = MOD); //needs to implement
void inverse(){
    for (int i = 0; i < MAX; ++i)
        inv[i] = bigmod(fact[i], MOD - 2);
}

int nCr(int a, int b){
    if (a < b or a < 0 or b < 0)
        return 0;
    int de = (inv[b] * inv[a - b]) % MOD;
    return (fact[a] * de) % MOD;
}

// nCr ends here
int ModInv(int a, int M){ return bigmod(a, M - 2, M);}
```

10 Some Algorithms**10.1 2D Prefix Sum** [2 lines] - 00be5300

```
pref[i][j] = a[i][j] + pref[i - 1][j] + pref[i][j - 1] -
    pref[i - 1][j - 1];
Sum of region = pref[row2][col2] - pref[row2][col1 - 1] -
    pref[row1 - 1][col2] + pref[row1 - 1][col1 - 1];
```

10.2 Big Integer [91 lines] - 9d0f0dfe

```
struct BigInteger {
    string str;
    // Constructor to initialize
    // BigInteger with a string
    BigInteger(string s) { str = s; }
    // Overload + operator to add
    // two BigInteger objects
    BigInteger operator+(const BigInteger& b) {
        string a = str, c = b.str;
        int alen = a.length(), clen = c.length();
        int n = max(alen, clen);
        if (alen > clen)
            c.insert(0, alen - clen, '0');
        else if (alen < clen)
            a.insert(0, clen - alen, '0');
        string res(n + 1, '0');
```

```

int carry = 0;
for (int i = n - 1; i >= 0; i--) {
    int digit = (a[i] - '0') + (c[i] - '0') + carry;
    carry = digit / 10;
    res[i + 1] = digit % 10 + '0';
}
if (carry == 1) {
    res[0] = '1';
    return BigInteger(res);
} else
    return BigInteger(res.substr(1));
}

```

// Overload - operator to subtract
// first check which number is greater and then subtract

```

BigInteger operator-(const BigInteger& b) {
    string a = str;
    string c = b.str;
    int alen = a.length(), clen = c.length();
    int n = max(alen, clen);
    if (alen > clen)
        c.insert(0, alen - clen, '0');
    else if (alen < clen)
        a.insert(0, clen - alen, '0');
    if (a < c) {
        swap(a, c);
        swap(alen, clen);
    }
    string res(n, '0');
    int carry = 0;
    for (int i = n - 1; i >= 0; i--) {
        int digit = (a[i] - '0') - (c[i] - '0') - carry;
        if (digit < 0)
            digit += 10, carry = 1;
        else
            carry = 0;
        res[i] = digit + '0';
    }
    // remove leading zeros
    int i = 0;
    while (i < n && res[i] == '0') i++;
    if (i == n) return BigInteger("0");
    return BigInteger(res.substr(i));
}

```

*// Overload * operator to multiply*
// two BigInteger objects

```

BigInteger operator*(const BigInteger& b) {
    string a = str, c = b.str;
    int alen = a.length(), clen = c.length();
    int n = alen + clen;
    string res(n, '0');
    for (int i = alen - 1; i >= 0; i--) {
        int carry = 0;
        for (int j = clen - 1; j >= 0; j--) {
            int digit = (a[i] - '0') * (c[j] - '0') +
                (res[i + j + 1] - '0') + carry;
            carry = digit / 10;
            res[i + j + 1] = digit % 10 + '0';
        }
        res[i] += carry;
    }
    int i = 0;
    while (i < n && res[i] == '0') i++;
    if (i == n) return BigInteger("0");
    return BigInteger(res.substr(i));
}

```

// Overload << operator to output
// BigInteger object

```

friend ostream& operator<<(ostream& out, const BigInteger&
    b) {
    out << b.str;
    return out;
}

```

```

}
};

```

10.3 Expression Parsing [92 lines] - c4a451ed

```

bool delim(char c) { return c == ' '; }

bool is_op(char c) { return c == '+' || c == '-' || c == '*'
    || c == '/'; }

bool is_unary(char c) { return c == '+' || c == '-'; }

int priority(char op) {
    if (op < 0) // unary operator
        return 3;
    if (op == '+' || op == '-') return 1;
    if (op == '*' || op == '/') return 2;
    return -1;
}

void process_op(stack<int>& st, char op) {
    if (op < 0) {
        int l = st.top();
        st.pop();
        switch (-op) {
            case '+':
                st.push(1);
                break;
            case '-':
                st.push(-1);
                break;
        }
    } else {
        int r = st.top();
        st.pop();
        int l = st.top();
        st.pop();
        switch (op) {
            case '+':
                st.push(1 + r);
                break;
            case '-':
                st.push(1 - r);
                break;
            case '*':
                st.push(1 * r);
                break;
            case '/':
                st.push(1 / r);
                break;
        }
    }
}

int evaluate(string& s) {
    stack<int> st;
    stack<char> op;
    bool may_be_unary = true;
    for (int i = 0; i < (int)s.size(); i++) {
        if (delim(s[i])) continue;

        if (s[i] == '(') {
            op.push('(');
            may_be_unary = true;
        } else if (s[i] == ')') {
            while (op.top() != '(') {
                process_op(st, op.top());
                op.pop();
            }
            op.pop();
            may_be_unary = false;
        } else if (is_op(s[i])) {
            char cur_op = s[i];
            if (may_be_unary && is_unary(cur_op)) cur_op = -cur_op;
            while (!op.empty() &&

```

```

                ((cur_op >= 0 && priority(op.top()) >=
                    priority(cur_op)) ||
                (cur_op < 0 && priority(op.top()) >
                    priority(cur_op)))) {
                process_op(st, op.top());
                op.pop();
            }
            op.push(cur_op);
            may_be_unary = true;
        } else {
            int number = 0;
            while (i < (int)s.size() && isalnum(s[i]))
                number = number * 10 + s[i++] - '0';
            --i;
            st.push(number);
            may_be_unary = false;
        }
    }

    while (!op.empty()) {
        process_op(st, op.top());
        op.pop();
    }
    return st.top();
}

```

10.4 Infix to Postfix [96 lines] - 92843ab9

```

bool delim(char c) { return c == ' '; }
bool is_op(char c) {
    return c == '+' || c == '-' || c == '*'
        || c == '/' || c == '^';
}
bool is_unary(char c) {
    return c == '+' || c == '-';
}
int priority(char op) {
    if (op < 0) return 3;
    if (op == '+' || op == '-') return 1;
    if (op == '*' || op == '/') return 2;
    if (op == '^') return 4;
    return -1;
}

void process_op(string& output, char op) {
    if (op < 0) {
        switch (-op) {
            case '+':
                output += "+ ";
                break;
            case '-':
                output += "- ";
                break;
        }
    }
    else {
        switch (op) {
            case '+':
                output += "+ ";
                break;
            case '-':
                output += "- ";
                break;
            case '*':
                output += "* ";
                break;
            case '/':
                output += "/ ";
                break;
            case '^':
                output += "^ ";
                break;
        }
    }
}

```

```

string InfixToPostFix(string& s) {
    string output;
    stack<char> op;
    bool may_be_unary = true;
    for (int i = 0; i < (int)s.size(); i++){
        if (delim(s[i]))
            continue;
        if (s[i] == '(') {
            op.push('(');
            may_be_unary = true;
        }
        else if (s[i] == ')') {
            while (op.top() != '(') {
                process_op(output, op.top());
                op.pop();
            }
            op.pop();
            may_be_unary = false;
        }
        else if (is_op(s[i])) {
            char cur_op = s[i];
            if (may_be_unary && is_unary(cur_op))
                cur_op = -cur_op;
            while (!op.empty() &&
                ((cur_op >= 0 && priority(op.top()) >=
                 priority(cur_op)) ||
                 (cur_op < 0 && priority(op.top()) >
                  priority(cur_op)))) {
                process_op(output, op.top());
                op.pop();
            }
            op.push(cur_op);
            may_be_unary = true;
        }
        else {
            char number;
            while (i < (int)s.size() && isalnum(s[i]))
                number = s[i++];
            --i;
            output.push_back(number);
            output.push_back(' ');
            may_be_unary = false;
        }
    }
    while (!op.empty()) {
        process_op(output, op.top());
        op.pop();
    }
    return output;
}

```

10.5 Kadanes [9 lines] - 6304cec2

```

// return maximum subarray sum.
int kadanes(int arr[], int n) {
    int maxsum = arr[0], curr_sum = arr[0];
    for (int i = 1; i < n; i++) {
        curr_sum = max(arr[i], curr_sum + arr[i]);
        maxsum = max(maxsum, curr_sum);
    }
    return maxsum;
}

```

10.6 Monotonic Stack (next greater) [7 lines] - b69c792c

```

for (int i = n - 1; i >= 0; i--) {
    while (!stk.empty() && v[i] >= v[stk.top()]) stk.pop();
    ind[i] = stk.empty() ? -1 : stk.top();
    stk.push(i);
}
// 3 1 5 4 10
// 2 2 4 4 -1

```

10.7 Spirally Traverse [25 lines] - 04cd0672

```

vector<int> spirallyTraverse(vector<vector<int>>& mat) {
    int m = mat.size();
    int n = mat[0].size();
}

```

```

vector<int> res;
vector<vector<bool>> vis(m, vector<bool>(n, false));
vector<int> dr = {0, 1, 0, -1};
vector<int> dc = {1, 0, -1, 0};
int r = 0, c = 0;
int idx = 0;
for (int i = 0; i < m * n; ++i) {
    res.push_back(mat[r][c]);
    vis[r][c] = true;
    int newR = r + dr[idx];
    int newC = c + dc[idx];
    if (0 <= newR && newR < m && 0 <= newC && newC < n &&
        !vis[newR][newC]) {
        r = newR;
        c = newC;
    } else {
        idx = (idx + 1) % 4;
        r += dr[idx];
        c += dc[idx];
    }
}
return res;
}

```

10.8 Ternary Search [10 lines] - d44f71ce

```

double ternary_search(double l, double r) {
    double eps = 1e-9; // error limit
    while (r - l > eps) {
        double m1 = l + (r - l) / 3, m2 = r - (r - l) / 3;
        double f1 = f(m1), f2 = f(m2); // evaluates the function
        at m1, m2
        if (f1 < f2) l = m1;
        else r = m2;
    }
    return f(l); // return the maximum of f(x) in [l, r]
}

```

11 String

11.1 AhoCorasick [78 lines] - 921b0745

```

struct AhoCorasick {
    struct Node {
        int next[26], fail = 0;
        int words = 0, out = 0;

        Node() {
            fill(next, next + 26, -1);
        }
    };

    vector<Node> trie;

    AhoCorasick() {
        trie.emplace_back();
    }

    // Insert pattern, return terminal node
    int insert(const string &s) {
        int node = 0;
        for (char ch : s) {
            int c = ch - 'a';
            if (trie[node].next[c] == -1) {
                trie[node].next[c] = trie.size();
                trie.emplace_back();
            }
            node = trie[node].next[c];
        }
        trie[node].words++;
        return node;
    }

    // Call once after inserting all patterns
    void build() {
        queue<int> q;
        trie[0].out = trie[0].words;
    }
}

```

```

for (int c = 0; c < 26; c++) {
    int ch = trie[0].next[c];
    if (ch == -1) {
        trie[0].next[c] = 0;
    } else {
        trie[ch].fail = 0;
        q.push(ch);
    }
}

while (!q.empty()) {
    int node = q.front();
    q.pop();

    int f = trie[node].fail;
    trie[node].out = trie[node].words + trie[f].out;

    for (int c = 0; c < 26; c++) {
        int ch = trie[node].next[c];

        if (ch == -1) {
            trie[node].next[c] = trie[f].next[c];
        } else {
            trie[ch].fail = trie[f].next[c];
            q.push(ch);
        }
    }
}

// ans[i] = number of patterns ending at index i
vector<int> search(const string &s) {
    vector<int> ans(s.size());
    int node = 0;

    for (int i = 0; i < (int)s.size(); i++) {
        node = trie[node].next[s[i] - 'a'];
        ans[i] = trie[node].out;
    }
    return ans;
}

};

11.2 Hashing [27 lines] - 6eal8b83
const ll H1=131, H2=137, M1=1e9+7, M2=1e9+9;
struct Hasher{
    int n; string s; vector<pll> h, pw;
    Hasher(const string&s): n(s.size()), s(s), h(n+1),
        pw(n+1){
        pw[0]={1,1};
        for(int i=1;i<=n;i++){
            pw[i]={pw[i-1].fi*H1%M1,pw[i-1].se*H2%M2};
            h[0]={0,0};
            for(int i=0;i<=n;i++){
                h[i+1]={(h[i].fi*H1+s[i])%M1,(h[i].se*H2+
                    s[i])%M2};
            }
        }
        pll get(int l,int r){
            ++r; // 0-indexed [l,r]
            return{(h[r].fi-h[l].fi*pw[r-l].fi%M1+M1*2)%M1,
                (h[r].se-h[l].se*pw[r-l].se%M2+M2*2)%M2};
        }

        int lcp(int i1,int i2,int n1,int n2){
            // LCP of s[i1..] and s[i2..]
            int lo=0,hi=min(n1,n2),ans=0;
            while(lo<=hi){
                int m=(lo+hi)/2;
                if(get(i1,i1+m-1)==get(i2,i2+m-1)){ans=m; lo=m+1;}
                else hi=m-1;
            }
            return ans;
        }
    };
};

```

11.3 KMP [69 lines] - e2b443a7

```

void constructLps(str &pat, vi &lps)
{
    ll len = 0;
    lps[0] = 0;
    ll i = 1;
    while (i < pat.length())
    {
        if (pat[i] == pat[len])
        {
            len++;
            lps[i] = len;
            i++;
        }
        else
        {
            if (len != 0)
            {
                len = lps[len - 1];
            }
            else
            {
                lps[i] = 0;
                i++;
            }
        }
    }
}

vi search(string &pat, string &txt, vi &lps)
{
    ll n = txt.length();
    ll m = pat.length();

    // vi lps(m);
    vi res;

    /// constructLps(pat, lps);

    ll i = 0;
    ll j = 0;

    while (i < n)
    {
        if (txt[i] == pat[j])
        {
            i++;
            j++;

            if (j == m)
            {
                res.push_back(i - j);
                j = lps[j - 1];
            }
        }
        else
        {
            if (j != 0)
                j = lps[j - 1];
            else
                i++;
        }
    }
    return res;
}

```

11.4 Suffix Array [168 lines] - f1d139d6

```

struct SuffixArray {
    vector<int> suffix_indices, suffix_rank, lcp;
    vector<vector<int>> sparse_table;

```

```

    SuffixArray(const string &s) {
        build(s);
    }

    // suffix_indices[i] = start of ith smallest suffix
    void build(const string &s) {
        suffix_indices.clear();
        suffix_rank.clear();
        lcp.clear();
        sparse_table.clear();

        int n = s.size();
        if (n == 0) return;

        string t = s + char(0);
        int m = t.size();

        vector<int> suffix_class(m), new_class(m);
        vector<int> new_indices(m), cnt(max(256, m), 0);
        suffix_indices.resize(m);

        for (unsigned char ch : t) cnt[ch]++;
        for (int i = 1; i < 256; i++)
            cnt[i] += cnt[i - 1];

        for (int i = 0; i < m; i++)
            suffix_indices[--cnt[(unsigned char)t[i]]] = i;

        suffix_class[suffix_indices[0]] = 0;
        int classes = 1;

        for (int i = 1; i < m; i++) {
            if (t[suffix_indices[i]] != t[suffix_indices[i - 1]])
                classes++;
            suffix_class[suffix_indices[i]] = classes - 1;
        }

        for (int h = 0; (1LL << h) < m; h++) {
            int len = 1 << h;

            for (int i = 0; i < m; i++) {
                new_indices[i] = suffix_indices[i] - len;
                if (new_indices[i] < 0)
                    new_indices[i] += m;
            }

            fill(cnt.begin(), cnt.begin() + classes, 0);

            for (int i = 0; i < m; i++)
                cnt[suffix_class[new_indices[i]]]++;

            for (int i = 1; i < classes; i++)
                cnt[i] += cnt[i - 1];

            for (int i = m - 1; i >= 0; i--)
                suffix_indices[--cnt[suffix_class[new_indices[i]]]]
                    = new_indices[i];

            new_class[suffix_indices[0]] = 0;
            classes = 1;

            for (int i = 1; i < m; i++) {
                pair<int, int> cur = {
                    suffix_class[suffix_indices[i]],
                    suffix_class[(suffix_indices[i] + len) % m]
                };

                pair<int, int> prev = {
                    suffix_class[suffix_indices[i - 1]],

```

```

                    suffix_class[(suffix_indices[i - 1] + len) % m]
                };

                if (cur != prev) classes++;
                new_class[suffix_indices[i]] = classes - 1;
            }

            suffix_class.swap(new_class);

            suffix_indices.erase(suffix_indices.begin());

            suffix_rank.resize(n);
            for (int i = 0; i < n; i++)
                suffix_rank[suffix_indices[i]] = i;

            build_lcp(s);

            // lcp[i] = LCP of suffix_indices[i] and [i+1]
            void build_lcp(const string &s) {
                int n = s.size(), k = 0;
                lcp.assign(max(0, n - 1), 0);

                for (int i = 0; i < n; i++) {
                    if (suffix_rank[i] == n - 1) {
                        k = 0;
                        continue;
                    }

                    int j = suffix_indices[suffix_rank[i] + 1];

                    while (i + k < n && j + k < n &&
                        s[i + k] == s[j + k])
                        k++;

                    lcp[suffix_rank[i]] = k;
                    if (k) k--;
                }

                // Call before get_lcp() or compare()
                void build_sparse_table() {
                    int n = lcp.size();
                    sparse_table.clear();
                    if (n == 0) return;

                    int levels = __lg(n) + 1;
                    sparse_table.assign(levels, vector<int>(n));
                    sparse_table[0] = lcp;

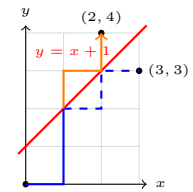
                    for (int k = 1; k < levels; k++) {
                        for (int i = 0; i + (1 << k) <= n; i++) {
                            sparse_table[k][i] = min(
                                sparse_table[k - 1][i],
                                sparse_table[k - 1][i + (1 << (k - 1))]
                            );
                        }
                    }

                    // LCP of suffixes starting at indices i and j
                    int get_lcp(int i, int j) {
                        int n = suffix_indices.size();
                        if (i == j) return n - i;

                        int left = suffix_rank[i];
                        int right = suffix_rank[j];
                        if (left > right) swap(left, right);

                        int k = __lg(right - left);
                        return min(
                            sparse_table[k][left],

```



- **Bad Path:** Touches $y = x + 1$.
- **Reflection:** Flip path after first touch.
- **Mapping:** Bad path to $(n, n) \iff$ Path to $(n-1, n+1)$.

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1} = \frac{(2n)!}{(n+1)!n!}$$

$$C_0 = 1, C_1 = 1, C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$$

• 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786

- Number of correct bracket sequences consisting of n opening brackets.

- Number of ways to completely parenthesize n+1 factors.

- The number of triangulations of a convex polygon with +2 sides (i.e. the number of partitions of polygon into disjoint triangles by using the diagonals).

- The number of ways to connect the 2n points on a circle to form n disjoint i.e. non-intersecting chords.

- The number of monotonic lattice paths from point (0,0) to point (n,n) in a square lattice of size $n \times n$, which do not pass above the main diagonal

- Number of permutation of length n that can be stack sorted.

- The number of non-crossing partitions of a set of n elements.

- The number of rooted full binary tree with n+1 leaves.

- The number of Dyck words of length 2n. A string consisting of n X's and n Y's such that no string prefix has more Y's than X's.

- Number of permutation of length n with no three-term increasing subsequence.

- Number of ways to tile a staircase shape of height n with n rectangle.

- $C_n^k = \frac{k+1}{n+1} \binom{2n-k}{n-k}$ denote the number of bracket sequences of size 2n with the first k elements being (.

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}$$

- The number of expressions containing n pairs of correct parentheses, which contain k distinct nestings. $N(4, 2) = 6$
 $()((()))$, $((()))()$, $((()()))$, $((()())())$, $((())())()$, $((())())()$

- The number of paths from (0,0) to (2n, 0) with steps only northeast and southeast, not staying below the x-axis with k peaks. And sum of all number of peaks is Catalan number.

```

        sparse_table[k][right - (1 << k)]
    );
}

// Compare s[l1..r1] and s[l2..r2]
// Returns -1 (smaller), 0 (equal), 1 (greater)
int compare(int l1, int r1, int l2, int r2) {
    int len1 = r1 - l1 + 1;
    int len2 = r2 - l2 + 1;

    int common = min({
        get_lcp(l1, l2), len1, len2
    });

    if (common == min(len1, len2)) {
        if (len1 == len2) return 0;
        return len1 < len2 ? -1 : 1;
    }

    return suffix_rank[l1] < suffix_rank[l2] ? -1 : 1;
};

```

11.5 SuffixAutomation [33 lines] - fee34b63

```

struct SAM {
    struct State { int len=0, link=0, nxt[26]={0}, fpos=0; ll
        cnt=0; };
    vector<State> st; int sz=2, last=1;
    SAM(int n) { st.resize(2*n+2); }
    // 1-Based indexing. 1 = empty string
    void add(char ch) {
        int c=ch-'a', cur=sz++, p=last, q, cl;
        st[cur].len=st[last].len+1; st[cur].cnt=1;
        st[cur].fpos=st[cur].len-1;
        for(; p && !st[p].nxt[c]; p=st[p].link)
            st[p].nxt[c]=cur;
        if(!p) st[cur].link=1;
        else if(st[p].len+1 == st[q=st[p].nxt[c]].len)
            st[cur].link=q;
        else {
            st[cl=sz++] = st[q]; st[cl].len=st[p].len+1;
            st[cl].cnt=0;
            for(; p && st[p].nxt[c]==q; p=st[p].link)
                st[p].nxt[c]=cl;
            st[q].link = st[cur].link = cl;
        }
        last=cur;
    }

    void build() {
        vector<int> o(sz-2); iota(all(o), 2);
        sort(all(o), [&](int a, int b){ return
            st[a].len<st[b].len; });
        for(int i=sz-3; i>0; i--) st[st[o[i]].link].cnt +=
            st[o[i]].cnt;
    }

    // --- Walk string in main ---
    // int u=1; for(char c: s) if(!(u=sam.st[u].nxt[c-'a']))
        break;
    // ll occ = u ? sam.st[u].cnt : 0; // Requires build()
    first
    // int first_start = u ? sam.st[u].fpos - s.length() + 1 :
        -1; // 0-indexed start pos

    // --- Distinct substrings ---
    // ll total=0; for(int i=2; i<sz; i++) total += st[i].len -
        st[st[i].link].len;
};

```

11.6 Trie [56 lines] - 105eac23

```

const int N = 26;
char BASE = 'A';
class Node {
public:

```

```

    int EoW;
    Node* child[N];
    Node() {
        EoW = 0;
        for (int i = 0; i < N; i++) child[i] = NULL;
    }
    Node* root = new Node();

    void insert(Node* node, string s) {
        for (size_t i = 0; i < s.size(); i++) {
            int r = s[i] - BASE;
            if (node->child[r] == NULL) node->child[r] = new Node();
            node = node->child[r];
        }
        node->EoW += 1;
    } // insert(root, s);

    int search(Node* node, string s) {
        for (size_t i = 0; i < s.size(); i++) {
            int r = s[i] - BASE;
            if (node->child[r] == NULL) return 0;
            node = node->child[r];
        }
        return node->EoW;
    } // search(root, s);

    void print(Node* node, string s = "") {
        if (node->EoW) cout << s << "\n";
        for (int i = 0; i < N; i++) {
            if (node->child[i] != NULL) {
                char c = i + BASE;
                print(node->child[i], s + c);
            }
        }
    } // print whole trie

    void remove(Node* node, string s) {
        for (size_t i = 0; i < s.size(); i++) {
            int r = s[i] - BASE;
            if (node->child[r] == NULL) return;
            node = node->child[r];
        }
        node->EoW--;
    } // remove(root, s)

    void delete_trie(Node* node) {
        for (int i = 0; i < N; i++) {
            if (node->child[i] != NULL) delete_trie(node->child[i]);
        }
        delete node;
    } // delete full trie

```

12 Random

12.1 Combinatorics

- $\sum_{k=0}^n \binom{n-k}{k} = Fib_{n+1}$
- $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$
- $k \binom{n}{k} = n \binom{n-1}{k-1}$
- Number of binary sequences of length n such that no two 0's are adjacent = Fib_{n+1}
- Number of non-negative solution of $x_1 + x_2 + x_3 + \dots + x_k = n$ is $\binom{n+k-1}{n}$

12.1.1 Catalan Number

- **Reflection Principle** ($n = 3$)
 Count paths $(0, 0) \rightarrow (n, n)$ below $y = x$.

12.1.2 Stirling Number of the First Kind

- Count permutation according to their number of cycles.
- $S(n, k)$ count the number of permutation of n elements with k disjoint cycles.
- $S(n, k) = (n-1) \times S(n-1, k) + S(n-1, k-1)$, $S(0, 0) = 1$, $S(n, 0) = S(0, n) = 0$
- $S(n, 1) = (n-1)!$
- $S(n, n-1) = \binom{n}{2}$
- $\sum_{k=0}^n S(n, k) = n!$

12.1.3 Stirling Numbers of the Second Kind

- Number of ways to partition a set of n objects into k non-empty subsets.
- $S(n, k) = k \cdot S(n-1, k) + S(n-1, k-1)$, $S(0, 0) = 1$, $S(n, 0) = S(0, n) = 0$
- $S(n, 2) = 2^{n-1} - 1$
- $S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$
- $S(n, k) * k! =$ number of ways to color n nodes using colors from 1 to k such that each color is used at least once.

12.1.4 Bell Number

- Counts the number of partitions of a set.
- $B_{n+1} = \sum_{k=0}^n \binom{n}{k} * B_k$
- $B_n = \sum_{k=0}^n S(n, k)$, where S is Stirling number of second kind.
- The number of multiplicative partitions of a square free number with i prime factors is the i -th Bell number.
- $B(p^m + n) \equiv mB(n) + B(n+1) \pmod{p}$
- If a deck is shuffled by removing and reinserting the top card n times, there are n^n possible shuffles. The number of shuffles that return the deck to its original order is B_n , so the probability of returning to the original order is B_n/n^n .

12.1.5 Lucas Theorem

- If p is prime then $\binom{p^a}{k} \equiv 0 \pmod{p}$
- For non-negative integers m and n and a prime p :

$$\binom{m}{n} = \prod_{i=0}^k \binom{m_i}{n_i} \pmod{p} \text{ where}$$

$$m = m_k p^k + m_{k-1} p^{k-1} + \dots + m_1 p + m_0 \quad n = n_k p^k + n_{k-1} p^{k-1} + \dots + n_1 p + n_0$$

are the base p expansion.

- Let n and k be non-negative integers, and let p be a prime. Let the base- p expansions of n and k be:

$$n = n_m p^m + n_{m-1} p^{m-1} + \dots + n_1 p^1 + n_0 p^0$$

$$k = k_m p^m + k_{m-1} p^{m-1} + \dots + k_1 p^1 + k_0 p^0$$

where $0 \leq n_i < p$ and $0 \leq k_i < p$ are the base- p digits of n and k , respectively.

The theorem states:

$$\binom{n}{k} \equiv \prod_{i=0}^m \binom{n_i}{k_i} \pmod{p}$$

Note: The individual terms $\binom{n_i}{k_i}$ are calculated using standard arithmetic, and the final product is taken modulo p .

- Example:** $\binom{10}{2} \pmod{2}$

$$n = 10_{10} = 1010_2$$

$$k = 2_{10} = 0010_2$$

Applying Lucas's Theorem:

$$\begin{aligned} \binom{10}{2} &\equiv \binom{1}{0} \cdot \binom{0}{0} \cdot \binom{1}{1} \cdot \binom{0}{0} \pmod{2} \\ &\equiv 1 \cdot 1 \cdot 1 \cdot 1 \equiv 1 \pmod{2} \end{aligned}$$

Bitwise check: $1010_2 \& 0010_2 = 0010_2$. Since $2 = 2$, the result is 1.

$$\binom{n}{k} \pmod{2} = \begin{cases} 1 & \text{if } (n \& k) = k \\ 0 & \text{otherwise} \end{cases}$$

12.1.6 Derangement

- A permutation such that no element appears in its original position.
- $d(n) = (n-1) * (d(n-1) + d(n-2))$, $d(0) = 1$, $d(1) = 0$
- $d(n) = nd(n-1) + (-1)^n = \lfloor \frac{n!}{e} \rfloor$, $n \geq 1$

12.1.7 Burnside Lemma

Given a group G of symmetries and a set X , the number of elements of X up to symmetry equals

$$\frac{1}{|G|} \sum_{g \in G} |X^g|$$

where X^g are the elements fixed by g ($g.x = x$) If $f(n)$ counts "configurations" of some sort of length n , we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n} \sum_{k=0}^{n-1} f(gcd(n, k)) = \frac{1}{n} \sum_{k|n} f(k) \phi(n/k)$$

12.1.8 Eulerian Number

- $E(n, k)$ is the number of permutations of the numbers 1 to n in which exactly k elements are greater than the previous element.
- $E(n, k) = (n-k)E(n-1, k-1) + (k+1)E(n-1, k)$, $E(n, 0) = E(n, n-1) = 1$
- $E(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$
- $E(n, k) = E(n, n-1-k)$
- $E(0, k) = [k=0]$
- $E(n, 1) = 2^n - n - 1$

12.2 Number Theory

12.2.1 Mobius Function and Inversion

- define $\mu(n)$ as the sum of the primitive n th roots of unity depending on the factorization of n into prime factors:

$$\mu(x) = \begin{cases} 0 & \text{n is not square free} \\ 1 & \text{n has even number of prime factors} \\ -1 & \text{n has odd number of prime factors} \end{cases}$$

- Mobius Inversion:

$$g(n) = \sum_{d|n} f(d) \leftrightarrow f(n) = \sum_{d|n} \mu(d) g(n/d)$$

- $\sum_{d|n} \mu(d) = [n=1]$
- $\phi(n) = \sum_{d|n} \mu(d) \cdot \frac{n}{d} = n \sum_{d|n} \frac{\mu(d)}{d} = \sum_{d|n} d \cdot \mu(\frac{n}{d})$
- $a|b \rightarrow \phi(a)|\phi(b)$
- $\phi(mn) = \phi(m) \cdot \phi(n) \cdot \frac{d}{\phi(d)}$ where $d = gcd(m, n)$
- $\phi(n^m) = n^{m-1} \phi(n)$
- $\sum_{i=1}^n [gcd(i, n) = k] = \phi(\frac{n}{k})$
- $\sum_{i=1}^n gcd(i, n) = \sum_{d|n} d \cdot \phi(\frac{n}{d})$
- $\sum_{i=1}^n \frac{1}{gcd(i, n)} = \sum_{d|n} \frac{1}{d} \cdot \phi(\frac{n}{d}) = \frac{1}{n} \sum_{d|n} d \cdot \phi(d)$
- $\sum_{i=1}^n \frac{i}{gcd(i, n)} = \frac{n}{2} \cdot \sum_{d|n} \frac{1}{d} \cdot \phi(\frac{n}{d}) = \frac{n}{2} \cdot \frac{1}{n} \sum_{d|n} d \cdot \phi(d)$
- $\sum_{i=1}^n \frac{n}{gcd(i, n)} = 2 \cdot \sum_{i=1}^n \frac{i}{gcd(i, n)} - 1$

12.2.2 GCD and LCM

- $gcd(a, b) = gcd(b, a \bmod b)$
- If $a|b.c$, and $gcd(a, b) = d$, then $(a/d)|c$.
- GCD is a multiplicative function.
- $gcd(a, lcm(b, c)) = lcm(gcd(a, b), gcd(a, c))$
- $gcd(n^a - 1, n^b - 1) = n^{gcd(a, b)} - 1$

12.2.3 Gauss Circle Theorem

- Determine the number of lattice points in a circle centered at the origin with radius r .
- number of pairs (m, n) such that $m^2 + n^2 \leq r^2$
- $N(r) = 1 + 4 \sum_{i=0}^{\infty} (\lfloor \frac{r^2}{4i+1} \rfloor - \lfloor \frac{r^2}{4i+3} \rfloor)$

- $\log_b b = 1$
- $\log_b 1 = 0$
- $\log_b x = \frac{\log_k x}{\log_k b}$

12.2.10 Miscellaneous

- $2^{100} = 2^{50} \times 2^{50}$
- Fibonacci Matrix:

$$\begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^{n-1} \begin{bmatrix} F_1 \\ F_0 \end{bmatrix}$$

- $\log(n!) = \log 1 + \log 2 + \dots + \log n$
- $\gcd(a, b) = \gcd(a - b, b)$
- Prime p in $n!$:

$$\left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \dots$$

- Catalan:

$$\binom{2n}{n} - \binom{2n}{n-1}$$

- Divisors of $p^x q^y$: $(x+1)(y+1)$
- Sum of divisors of $p^x q^y$:

$$(1 + p + p^2 + \dots + p^x)(1 + q + q^2 + \dots + q^y)$$

- Divisibility rules:

- 3: Sum of digits divisible by 3
- 4: Last two digits divisible by 4
- 5: Last digit 0 or 5
- 6: Divisible by both 2 and 3
- 7: Double last digit, subtract from rest. Continue till manageable.
- 8: Last three digits divisible by 8
- 9: Sum of digits divisible by 9
- 11: $(\text{sum of odd digits}) - (\text{sum of even digits}) \equiv 0 \pmod{11}$

- Derangement: $D_n = (n-1)D_{n-2} + (n-1)D_{n-1}$
- Golden Ratio: $\Phi = \frac{1+\sqrt{5}}{2} \approx 1.618034$
- n -th Fibonacci:

$$F_n = \frac{\Phi^n - (1-\Phi)^n}{\sqrt{5}}$$

- Stars & Bars: Solutions of $x_1 + x_2 + \dots + x_k = n$ $x_i > 0$: $\binom{n-1}{k-1}$, $x_i \geq 0$: $\binom{n+k-1}{k-1}$
- Throwing Dice (Matrix Exponentiation)

- **Recurrence Relation** The number of ways to achieve a sum n is given by the linear recurrence:

$$f(n) = f(n-1) + f(n-2) + f(n-3) + f(n-4) + f(n-5) + f(n-6)$$

with the base case $f(0) = 1$ and $f(k) = 0$ for $k < 0$.

- **State Transition** The relationship between the new state vector (V_i) and the old state vector (V_{i-1}) is:

$$\underbrace{\begin{bmatrix} f(i) \\ f(i-1) \\ f(i-2) \\ f(i-3) \\ f(i-4) \\ f(i-5) \end{bmatrix}}_{V_i} = \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}}_T \times \underbrace{\begin{bmatrix} f(i-1) \\ f(i-2) \\ f(i-3) \\ f(i-4) \\ f(i-5) \\ f(i-6) \end{bmatrix}}_{V_{i-1}}$$

- **Final Answer** The n -th term is found by: $f(n) = (T^n)_{1,1} \times f(0)$, where $f(0) = 1$.

$$f(n) = (T^n)_{1,1}$$

12.3 Probability and Expectation

- Let X be a discrete random variable with probability $p_X(x)$ of assuming the value x . It will then have an expected value (mean) $\mu = \mathbb{E}(X) = \sum_x x p_X(x)$ and variance $\sigma^2 = V(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_x (x - \mathbb{E}(X))^2 p_X(x)$ where σ is the standard deviation. If X is instead continuous it will have a probability density function $f_X(x)$ and the sums above will instead be integrals with $p_X(x)$ replaced by $f_X(x)$.
- Expectation is linear:

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

For independent X and Y ,

$$V(aX + bY) = a^2 V(X) + b^2 V(Y).$$

- **Discrete distributions**

- **Binomial distribution:** The number of successes in n independent yes/no experiments, each which yields success with probability p is $\text{Bin}(n, p)$, $n = 1, 2, \dots$, $0 \leq p \leq 1$.

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mu = np, \sigma^2 = np(1-p)$$

$\text{Bin}(n, p)$ is approximately $\text{Po}(np)$ for small p .

- **First success distribution:** The number of trials needed to get the first success in independent yes/no experiments, each which yields success with probability p is $\text{Fs}(p)$, $0 \leq p \leq 1$.

$$p(k) = p(1-p)^{k-1}, k = 1, 2, \dots$$

$$\mu = \frac{1}{p}, \sigma^2 = \frac{1-p}{p^2}$$

- **Poisson distribution:** The number of events occurring in a fixed period of time t if these events occur with a known average rate κ and independently of the time since the last event is $\text{Po}(\lambda)$, $\lambda = t\kappa$.

$$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, k = 0, 1, 2, \dots$$

$$\mu = \lambda, \sigma^2 = \lambda$$

- **Continuous distributions**

- **Uniform distribution:** If the probability density function is constant between a and b and 0 elsewhere it is $\text{U}(a, b)$, $a < b$.

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

$$\mu = \frac{a+b}{2}, \sigma^2 = \frac{(b-a)^2}{12}$$

- **Exponential distribution:** The time between events in a Poisson process is $\text{Exp}(\lambda)$, $\lambda > 0$.

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\mu = \frac{1}{\lambda}, \sigma^2 = \frac{1}{\lambda^2}$$

- **Normal distribution:** Most real random values with mean μ and variance σ^2 are well described by $\mathcal{N}(\mu, \sigma^2)$, $\sigma > 0$.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ then

$$aX_1 + bX_2 + c \sim \mathcal{N}(\mu_1 + \mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2)$$